ON GRAFFITI/WOWII CONJECTURE 133: A PROVED REDUCTION, THE RANGE $\ell \le 3$, AND QUARANTINED COMPUTER RECORDS

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Abstract. Conjecture 133 of *Written on the Wall II*, generated by the program Graffiti.pc, states that every connected graph G on at least two vertices with no 4-cycle subgraph satisfies $p(G) \ge r(G) + \ell(G)$. Here $p(G)$ is the largest order of an induced path, $r(G)$ is the radius, and $\ell(G) = S(G)/n$ is the average over the vertices v of $\alpha(G[N(v)])$. We call this statement REAL-133. We prove it when $\ell(G) \le 3$, when $r(G) \le 2$ and $\ell(G) \le 4$, and when $r(G) \ge 5$ and $\ell(G) \le 4$. A theorem of Erdős, Saks and Sós reduces a counterexample to one of two families: a radius-free mass bound GEN_K, or a radius-extremal bound $\mathrm{RAD}(r)$. That reduction is proved. We also prove the rooted bound ROOT(3), and we prove ROOT(q) for every q in the class of connected strong C_4-free graphs whose blocks are bipartite or cycles. Several natural strengthenings fail, by explicit examples written in the text.

The following bounds are not theorems of this paper. Each still depends on an enumeration, a supplement, or a search payload that is not in this archive, and each is marked not independently checkable from this archive: GEN_3, GEN_4, ROOT(q) for $4 \le q \le 8$, RAD(r) for $4 \le r \le 10$, the bipartite form of GEN_5, and the structure statement Struct(r). A written deduction would give REAL-133 for $\ell \le 5$ if those inputs were theorems. They are not, so REAL-133 for $3 < \ell \le 5$ is not claimed. Partial Result G is not a theorem. Three reductions attached to it are proved: a deletion identity, a degeneracy implication under an explicit path hypothesis, and a conditional equivalence. The mass and deletion statements that would be needed to empty its residual class $\mathcal{R}_G$ are verdict records from rounds R35–R40; the searches behind those records are absent, and $\mathcal{R}_G$ is not shown to be empty. Appendix E proves two port lemmas and an averaging inequality, gives one counterexample to a one-vertex quotient, and quarantines every further lemma whose justification is a missing computation. Unrestricted GEN_6, RAD(r) for $r \ge 11$, and REAL-133 remain open.

1. Introduction

All graphs are finite and simple. For a graph G with $n \ge 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius (measured in edges), and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where α is the independence number and $N(v)$ the open neighbourhood. Write $c_{C_4}(G) = 1$ if G contains no 4-cycle as a subgraph (chorded or not) and $c_{C_4}(G) = 0$ otherwise.

Conjecture 1.1 (Graffiti/WOWII Conjecture 133)**.** *Every connected graph G on at least two vertices satisfies*

$$r(G) + \ell(G)^{\,c_{C_4}(G)} \le p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* of conjectures of the program Graffiti.pc [3]. We follow the formal statement in the Formal Conjectures repository [5], file `FormalConjectures/WrittenOnTheWallII/GraphConjecture133.lean`, which reads `(rad : ℝ) + l G ^ cC4 ≤ (path G : ℝ)`. Earlier versions of that file had $\lfloor \ell(G) \rfloor$ in place of $\ell(G)$. Pull request #5727 of the repository [6], "drop the floor on `l G`" (merged on 13 September 2026),

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removed the floor on the grounds that the brackets in the WOWII source denote grouping. We work with the real-valued statement throughout and call it REAL-133.

If G contains a 4-cycle, the conjecture reads $p(G) \geq r(G)+1$, and this always holds: a shortest path between two vertices at distance $\operatorname{diam}(G) \geq r(G)$ is induced. So only graphs without a 4-cycle subgraph matter. We call them *strong C_4-free*; the word "strong" emphasises that chorded 4-cycles are excluded too. Since p and r are integers, for a strong C_4-free G the conjecture is equivalent to

$$p(G) - r(G) \;\geq\; \lceil \ell(G) \rceil. \tag{1}$$

H. Chen [2] has studied mainly the earlier floor version $p \geq r + \lfloor \ell \rfloor$. That preprint describes itself as AI-assisted and not peer-reviewed, and we have not checked its proofs. For the floor version it claims proofs when $\lfloor \ell \rfloor \leq 2$, when $\lfloor \ell \rfloor = 3$ and $r = 2$, and for minimum-order counterexamples with $\lfloor \ell \rfloor = 3$ and $r = 3$. For the real-valued form it records an exhaustive check on strong C_4-free graphs with at most 13 vertices, and its Theorem 6.7 states that $\operatorname{diam}(G) = 2$ and $\ell > 3$ imply $p \geq 6$; this overlaps with the case $r = 2$, $3 < \ell \leq 4$ of our Theorem A for graphs of diameter 2. Its Lemma 7.18 contains the trace classification in the first part of our Lemma 3.4. Our proofs are independent of [2].

Status labels. Every statement carries one of the following labels.

- **Proved**: a complete written proof in this paper, possibly using the published theorems quoted in Section 2.
- **Proved (written proof in the supplement)**: a complete written proof without computer assistance that is too long to reproduce here; the exact file is named in the statement and in Appendix F. If that file is not in this repository, the proof is not checkable here (Section 16).
- **Finite classification**: a written proof that also uses an exhaustive classification of small base configurations (at most 13 vertices; single-vertex attachment tests on them have at most 14 vertices), intended to be re-run by a stand-alone checker. That checker is not in this repository (Section 16).
- **Computer-assisted, payload absent**: a written reduction together with a finite enumeration that is not a file in this repository. Such a statement is quarantined. It is not a theorem of this paper.
- **Quarantined (not independently checkable from this archive)**: the mathematical sentence is printed, together with the missing input. It is not part of the citable claim ladder. No proved theorem takes it as a hypothesis. A conditional sentence may name it as an explicit supposition; the conclusion is then not claimed either. A certificate hash authenticates stored JSON bytes only. It does not replace a missing payload.
- **Computer-assisted partial result**: proved reductions, which are theorems, kept separate from verdict records, which are not. Any residual class is part of the statement and is not shown to be empty. Such a result does not close an open problem of Section 13.
- **Counterexample**: an explicit witness, proved in the text, that a stated construction fails or that a stated universal sentence is false, at the quantifiers written in the lemma. The witness is not used as a positive hypothesis elsewhere.
- **Formalised**: the statement and its proof are checked by the Lean 4 kernel. The Lean development named in Section 5.3 is not included in this repository.
- **Conjecture** or **open**: not proved. Finite evidence, if any, is stated with its exact domain.

The reduction. Write $S(G) = \sum_v \alpha(G[N(v)]) = n\ell(G)$. For integers $K \geq 3$ and $r \geq 4$ consider:
GEN_K: every connected strong C_4-free graph G with $p(G) \leq 2K$ satisfies $S(G) \leq Kn$;

RAD(r): every connected strong C_4-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \leq r - 1$.

Theorem B (proved; Theorem 4.1)**.** *Let G be a counterexample to REAL-133. Then $\ell(G) > 3$, and with $K = \lceil \ell(G) \rceil - 1 \geq 3$, either $r(G) \leq K$ and $p(G) \leq 2K$ (so G violates* GEN_K*), or $r(G) = K + 1$ and $p(G) = 2r(G) - 1$ (so G violates* $\mathrm{RAD}(K + 1)$*).*

Main results.

Theorem A (proved; Theorem 3.7)**.** *REAL-133 holds for every connected strong C_4-free graph G with $\ell(G) \leq 3$, and for every one with $r(G) \leq 2$ and $\ell(G) \leq 4$.*

For a vertex s of a connected graph F let $p_s(F)$ be the largest order of an induced path starting at s, $D = d_F(s)$, and $m = e(F[N(s)])$. The rooted bound ROOT(q) asks that every connected strong C_4-free F with a vertex s such that $p_s(F) \leq q$ satisfy $S(F) \leq (q-1)(|F|-1)-(q-3)D-m$. Lemma 5.2 proves ROOT(3). Parts (1) and (2) of Theorem 9.1 prove ROOT(q) for every q when every block is bipartite or a cycle.

Section 11 collects negative results, proved from explicit graphs: star bounds fail for $K = 5$; an endpoint inequality fails at $q = 9$; a dominating-core budget is refuted by the Petersen graph; and one scalar system satisfied by the numerical data of the Heawood graph does not imply REAL-133. Grafts of projective-plane incidence graphs show that ℓ is unbounded on the radius-extremal class $p = 2r - 1$, so RAD(r) must depend on r.

Claim ladder. The citable theorems of this paper are Theorem 3.7, Theorem 4.1, Corollary 7.3, Lemma 5.2, parts (1) and (2) of Theorem 9.1, Proposition 9.3, the examples of Section 11, and the three reductions in Section 1, together with the port lemmas and the averaging inequality of Appendix E. Everything else in the introduction is quarantined.

Quarantine: bounds that are not independently checkable. No statement in this subsection is a theorem. Each one still depends on an enumeration, a supplement, or a search payload that is not a file in this repository (Section 16). A matching certificate hash authenticates stored JSON. It does not supply the missing file, and it does not place the inequality on the claim ladder.

Quarantined record (GEN_3 and GEN_4)**.** The inequalities $S \leq 3n$ for strong C_4-free P_7-free graphs and $S \leq 4n$ for connected strong C_4-free P_9-free graphs are printed later as Quarantined statements 5.1 and 6.1. The first depends on three finite classifications whose configuration lists and checker programs are absent; the Lean 4 development of Section 5.3 is absent as well. The second depends on an extension DAG that is absent. Neither inequality is a hypothesis of a proved theorem.

Quarantined record (rooted bounds for $q \geq 4$, and RAD(r) for $4 \leq r \leq 10$)**.** ROOT(q) for $4 \leq q \leq 8$ and RAD(r) for $4 \leq r \leq 10$, including the shape $S \leq (r-1)n-2r(r-3)$ for $5 \leq r \leq 10$, are printed in Sections 7 and 8. For $q = 4$ and $q = 5$ the written proofs lie in supplementary reports that are not in this repository. For $6 \leq q \leq 8$ and for $5 \leq r \leq 10$ the arguments use enumerations that are not in this repository. The radius-four bound uses a classification of configurations on at most 13 vertices whose list is not in this repository. None of these bounds is a theorem of this paper.

Quarantined record (the range $\ell \leq 5$; not claimed)**.** If GEN_3, GEN_4, the radius-four bound and the radius-five bound were theorems, the deduction written under Quarantined statement 8.3 would give REAL-133 for every connected graph with $\ell \leq 5$. Those four inputs are quarantined. REAL-133 for $3 < \ell \leq 5$ is not a theorem of this paper.

Quarantined record (bipartite GEN_5 and the radius form in the block class)**.** Part (3) of Theorem 9.1 uses Struct(r), whose proof is not in this repository. The bipartite bound of Quarantined statement 9.5 uses an extension certificate that is not in this repository. Bipartite GEN_5 is therefore not a theorem of this paper. General GEN_5 is open.

Partial Result G. Let $\mathcal{F}$ be the class of finite triangle-free strong C_4-free graphs with $p \le 12$. Write $\varepsilon_6(G) = S(G) - 6|G|$, and call G *positive* when $\varepsilon_6(G) > 0$. A nonempty set $U \subseteq V(G)$ is 6-*cheap* when $D_G(U) \le 6|U|$.

Partial Result G is not a theorem. It is the pair consisting of the proved reductions below and the quarantined verdict records that follow them. The two parts are not interchangeable. The records do not empty the residual class $\mathcal{R}_G$ defined after them. They do not prove unrestricted triangle-free GEN_6, and they do not prove REAL-133.

Proved reductions.

Lemma 1.2 (deletion margin; proved)**.** *Let G be triangle-free and strong C_4-free, and let $\varnothing \neq U \subseteq V(G)$. Then*

$$D_G(U) = 2e\big(G[U]\big) + 2e(U, V(G) \setminus U),$$

and $\varepsilon_6(G) - \varepsilon_6(G - U) = D_G(U) - 6|U|$. If $D_G(U) \le 6|U|$ and $\varepsilon_6(G) > 0$, then $U \neq V(G)$ and $\varepsilon_6(G - U) \ge \varepsilon_6(G) > 0$. If $K \ge 6$ is real and $D_G(U) \le 6|U|$, then $D_G(U) \le K|U|$.

Proof. By Lemma 2.1, $\lambda(v) = d(v)$ and $S(G) = 2e(G)$. The edges counted by $S(G)$ but not by $S(G-U)$ are exactly the edges that meet U, so $D_G(U) = 2e(G[U]) + 2e(U, V(G)\setminus U)$. The identity for ε_6 is then the definition. If $U = V(G)$, then $D_G(U) = S(G)$, hence $D_G(U) - 6|U| = \varepsilon_6(G)$. Thus $\varepsilon_6(G) > 0$ forbids $D_G(V(G)) \le 6|V(G)|$. The inequality $D_G(U) \le 6|U|$ rearranges to $\varepsilon_6(G - U) \ge \varepsilon_6(G)$. For real $K \ge 6$ one has $6|U| \le K|U|$. □

Lemma 1.3 (degeneracy implication; proved from an explicit hypothesis)**.** *Assume that every nonempty triangle-free strong C_4-free graph of minimum degree at least 4 contains an induced P_{11}. Then every triangle-free strong C_4-free graph with $p \le 10$ is 3-degenerate, and therefore satisfies $e \le 3(n-1)$ and $S \le 6(n-1)$ whenever $n \ge 1$. In particular $S \le 6n$.*

Proof. Let H be a nonempty induced subgraph of a triangle-free strong C_4-free graph with $p \le 10$. Then H is triangle-free and strong C_4-free, and $p(H) \le 10$, so H has no induced P_{11}. The assumed hypothesis forces $\delta(H) \le 3$. Delete a vertex of degree at most 3 in the current graph and repeat. Each of the first $n-1$ deletions removes at most three edges, and the last vertex is isolated, so $e \le 3(n-1)$. Lemma 2.1(3) gives $S = 2e$. □

The path hypothesis of Lemma 1.3 is not proved in this paper. The same sentence appears among the verdict records below. The lemma adds only the peeling bound, and only under that hypothesis.

Proposition 1.4 (equivalence under an explicit supposition; proved)**.** *Let $\mathcal{F}$ be the class defined above, and let $\mathcal{R} \subseteq \mathcal{F}$. Suppose that every positive graph in $\mathcal{F} \setminus \mathcal{R}$ has a nonempty 6-cheap set. Then the following are equivalent.*

1. *Every positive graph in $\mathcal{R}$ has a nonempty 6-cheap set.*
2. *Every graph in $\mathcal{F}$ satisfies $S \le 6n$.*

The supposition is not a proof that $\mathcal{R}$ is empty, and neither (1) *nor* (2) *is proved in this article.*

Proof. If (2) holds, then no graph in $\mathcal{F}$ is positive, so (1) is vacuous. Conversely, assume (1) and the supposition, and suppose some graph in $\mathcal{F}$ is positive. Among all positive induced subgraphs of members of $\mathcal{F}$, choose H of minimum order. Then $H \in \mathcal{F}$. A positive component of H would be a positive induced subgraph, so H is connected. If H had a nonempty 6-cheap set U, Lemma 1.2 would give a smaller positive induced subgraph $H - U$, a contradiction. Thus $H \notin \mathcal{F} \setminus \mathcal{R}$, so $H \in \mathcal{R}$, contradicting (1). Hence no graph in $\mathcal{F}$ is positive, which is (2). □

Enumerated verdict records, not theorems. The three lists below are not proved. Each inequality is supported in this repository only by a verdict mark in a JSON file from rounds R35–R40. The completion DAGs, trace catalogues and checker programs named in those files are absent. The lists are printed so that the residual class can be named. They establish no bound, and they are not cited by any proved theorem.

Quarantined record (records (M), (S) and (P); residual $\mathcal{R}_G$ open)**.** (1) **Mass sentences (M), not theorems.**

(a) For every triangle-free strong C_4-free graph with $p \leq 11$, the sentence $S \leq 6n$ is a verdict record. The recorded argument takes general GEN_5 for $p \leq 10$ as a premise. General GEN_5 is open. The same numerical bound for $p \leq 10$ is the conclusion of Lemma 1.3 under a path hypothesis that is itself only a verdict record.

(b) For $p \leq 11$ and $n \geq 8$, the sentence $S \leq 6n - 28$ is a verdict record.

(c) For $p \leq 12$, $n \geq 14$, and every endpoint of an induced P_{12} of degree at most 4, the sentence $S \leq 6n - 36$ is a verdict record.

(d) For graphs in $\mathcal{F}$, the sentence that an endpoint-reducing deletion of weight at most 18 which leaves at least 14 vertices implies $S \leq 6n$ is a verdict record.

(e) For $p \leq 12$ and at most 31 vertices of degree at least 5, and again with 31 replaced by 32 and with no bound on the remaining degrees, the sentence $S \leq 6n$ is a verdict record. If in addition every endpoint of an induced P_{12} has degree at most 5, the same sentence with 31 and 32 replaced by 35 and 36 is a verdict record.

The contact-catalogue steps inside these arguments are not separate theorems. Their search payloads are absent.

(2) **Structural sentences (S), not theorems.**

(a) The sentence that every nonempty triangle-free strong C_4-free graph of minimum degree at least 4 contains an induced P_{11} is a verdict record. It is the hypothesis of Lemma 1.3, not a theorem.

(b) There is a recorded triangle-free strong C_4-free graph on 112 vertices and 337 edges with $\varepsilon_6 = 2$, with 29 vertices of degree at least 5, and with 112 rooted induced paths on 13 vertices. The same record names 29 as a minimum, inside its own class, for the number of vertices of degree at least 5 among graphs with no 6-cheap set. The edge list of this graph is not in the paper, so the sentence is not checkable here. A second record counts 106 omission allocations for a graph denoted C_{106} in that file. Those allocations are not restated.

(3) **Robertson punctures (P), not theorems.** The following sentence is a verdict record, not a classification proved in this paper. A connected graph of minimum degree at least 4 that contains a one-vertex puncture of the Robertson graph, in the sense used by that record, is isomorphic to one of four graphs denoted in the record by R, X_{21}, X_{22} and X_{23}. Whenever such a puncture is induced, the record names a nonempty 6-cheap set of order at most 7, and names order 7 as sharp on X_{22}. The record also says that every positive graph in $\mathcal{F}$ reduces, by deleting such sets, to a positive induced subgraph containing none of X_{21}, X_{22} and X_{23}. The isomorphism types are not reproduced here, and the search is absent, so this sentence is not a theorem.

(4) **Residual class.** Let $\mathcal{R}_G$ be the class of positive graphs in $\mathcal{F}$ for which none of the cheap-deletion sentences in (M) and (P) supplies a nonempty 6-cheap set. Nothing above proves $\mathcal{R}_G = \varnothing$. In particular nothing above proves that every triangle-free strong C_4-free graph with $p \in \{11, 12\}$, minimum degree at least 4, and every endpoint of every longest induced path of degree at most 7, has a nonempty 6-cheap set. Two further verdict records state a conditional equivalence, relative to exclusions accepted in those sessions, between cheap deletion on their residual, the endpoint-bounded deletion problem of Section 13,

and $S \le 6n$ on $\mathcal{F}$. The case division behind that relative hypothesis is not in this paper. Proposition 1.4 is the equivalence that is proved, and it assumes its supposition rather than proving it. Unrestricted triangle-free GEN_6 is open. REAL-133 is open.

Appendix E does not remove $\mathcal{R}_G$. Its proved lemmas do not assume that every graph in $\mathcal{F}$ has a 6-cheap set. Its quarantined notes are not theorems. Lemma E.5 is a counterexample to one construction.

Statement	Proved in text	Computer-recorded	Payload absent	Open residual
Thms. 3.7, 7.3	yes	no	no	no
Theorem 4.1	yes	no	no	no
ROOT(3)	yes	no	no	no
Block class, ROOT(q) all q	yes, Thm. 9.1(1)(2)	no	no	no
Regular bipartite; Moore	yes, Prop. 9.3	no	no	no
Deletion, ports, average	yes, Lems. 1.2, E.1, E.2, E.4	no	no	no
Equivalence	yes, under its supposition	no	no	does not empty $\mathcal{R}$
GEN_3	reduction only	three finite counts	programs and Lean absent	not a theorem
GEN_4	outline only	verdict record	extension DAG absent	not a theorem
RAD(4)	trace argument after the core	finite list	list absent	not a theorem
ROOT(4), ROOT(5)	no	no	supplement absent	not a theorem
ROOT(6)–8, RAD(5)–10	outline only	verdict records	searches absent	not a theorem
REAL-133 for $\ell \le 5$	deduction only	inherits the previous rows	yes	not claimed
Struct(r); bipartite GEN_5	block identities (1)(2) only	verdict records	base and DAG absent	not a theorem
Partial Result G, records (M)(S)(P)	no	R35–R40 verdicts	searches absent	$\mathcal{R}_G$ open
Other notes, App. E	one graph	R51–R53	R45 absent	not GEN_6
Unrestricted GEN_6; REAL-133	no	no	not supplied	open

TABLE 1. Dependencies. A row is on the claim ladder only when "Proved in text" is yes and the other three columns do not withhold the statement. "Payload absent" means the enumeration, supplement, or checker required by that row is not in this repository.

2. Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing v. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

Lemma 2.1 (proved)**.** *Let G be strong C_4-free. Then:*

(1) *two distinct vertices have at most one common neighbour;*
(2) *for every v, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*

(3) $S(G) = 2e(G) - 3T(G)$.

Proof. (1) Two common neighbours w, w' of x, y give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with v. So the components are single vertices and edges, each edge being a triangle at v. (3) Sum $d(v) - t(v)$ over v. □

The components of $G[N(v)]$ are the *units* of v; each is a vertex or an edge. For a vertex set U, $G - U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of U. We write $L(v) = D_G(\{v\})$.

Lemma 2.2 (deletion loss; proved)**.** *Let G be strong C_4-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.

Proof. For $x \notin U$, deleting U removes exactly the units of x that lie inside U; a unit that is an edge with one end in U survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of v in triangles with v keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

Lemma 2.3 (cut vertices; proved)**.** *Let v be a cut vertex of G. Split the components of $G - v$ into two nonempty groups, and let G_1, G_2 be the subgraphs induced by the two groups together with v. Then $S(G) = S(G_1) + S(G_2)$.*

Proof. $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

Theorem 2.4 (Erdős, Saks and Sós [4])**.** *Every connected graph G has an induced path on at least $2r(G) - 1$ vertices.*

Theorem 2.5 (Camby and Schaudt [1, Theorem 5])**.** *Let $k \geq 4$ and let G be a connected graph with no induced path on k vertices. Then G has a connected dominating set X such that $G[X]$ has no induced path on $k - 2$ vertices or $G[X]$ is isomorphic to the cycle C_k.*

Theorem 2.6 (Seinsche [10])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [1, Theorem 5] applied with $H = G$. We write P_k for the path on k vertices, and "G is P_k-free" means that G has no induced P_k, that is, $p(G) \leq k - 1$. We also use the elementary fact that a connected graph of radius r has at least $2r$ vertices (a spanning tree has radius at least r, and a tree on n vertices has radius at most $\lfloor n/2 \rfloor$).

Lemma 2.7 (proved)**.** *Let H be a connected strong C_4-free P_4-free graph on at least two vertices. Then H has a vertex adjacent to all other vertices.*

Proof. By Theorem 2.6, H is the join of two nonempty vertex sets A and B. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

3. Small mass: the range $\ell \leq 3$

Lemma 3.1 (radius one; proved)**.** *If G is connected and strong C_4-free with $r(G) = 1$, then $S(G) = 2(n - 1) - m$, where m is the number of edges of $G[N(u)]$ for a dominating vertex u. Hence $\ell(G) < 2$, and (1) holds.*

Proof. Let $v \neq u$. By Lemma 2.1(1), v has at most one neighbour other than u, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \leq 2$. If G is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with u form an induced P_3, so $p \geq 3 \geq r + \lceil \ell \rceil$. □

Lemma 3.2 (proved)**.** *If G is connected and strong C_4-free with $r(G) \geq 2$, then $p(G) \geq 4$.*

Proof. If G were P_4-free, Lemma 2.7 would give a dominating vertex, so $r(G) \leq 1$. □

Lemma 3.3 (P_5-free mass; proved)**.** *If G is connected, strong C_4-free and P_5-free, then $S(G) \leq 2n - h$, where h is the number of vertices v with $\lambda(v) \geq 3$. In particular $S(G) \leq 2n$.*

Proof. Call v *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \geq 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of x if x lies in an edge unit of v.

Key fact. Let x, y lie in different units of v. A common vertex of X_x and X_y would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced P_5: $x \not\sim y$ because they lie in different units, $x \not\sim y'$ and $y \not\sim x'$ by disjointness, and v is adjacent to neither x' nor y'. So X_x is complete to X_y. If both are nonempty and $|X_x| \geq 2$, a vertex of X_y has two neighbours in X_x, which are also neighbours of x, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.

Main step. For each high v we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is v, one in each of $\lambda(v) - 1$ distinct units of v.

Case 1: some $u \in N(v)$ has $|X_u| \geq 2$. By the key fact $X_y = \emptyset$ for every y outside the unit of u. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so y is simplicial; so is $\bar{y}$. Hence v is the only high neighbour of y, and the $\lambda(v) - 1$ units other than that of u supply the vertices.

Case 2: $|X_x| \leq 1$ for all $x \in N(v)$. Then $\lambda(x) \leq 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices x has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let x lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced P_5: $x \not\sim y$ (different units), $x \not\sim y', z'$ (disjointness), $v \not\sim y', z'$, and $y \not\sim z'$ (disjointness). As $\lambda(v) \geq 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is v. This gives at least $\lambda(v) - 1$ units.

Counting. The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}} (\lambda(v) - 1) \leq \#\{\text{simplicial vertices}\}$, and

$$S = \sum_v \lambda(v) \leq 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v) - 2) \leq 2n - h.$$ □

Lemma 3.4 (C_6 attachments; proved)**.** *Let G be connected, strong C_4-free and P_6-free with a dominating induced cycle $C = v_0 v_1 \cdots v_5$, and let $W = V(G) \setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of C, distinct vertices of W use distinct pairs, and W is independent. Hence $|W| \leq 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \leq 2$ if $W \neq \emptyset$.*

Proof. Indices are modulo 6. Let $w \in W$; it has a neighbour on C. If w has exactly one neighbour v_i, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced P_6. If $N(w) \cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced P_6. Two neighbours at distance 2 on C give a 4-cycle. A set of at least three vertices of C_6 contains two vertices at distance 2. So $N(w) \cap C$ is an antipodal pair. Two vertices of W with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on C, and wv_iv_jw' is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if w uses $\{v_i, v_{i+3}\}$, then every vertex of C is within distance 2 of v_i (the vertex v_{i+3} through w), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So v_i has eccentricity 2. □

A stand-alone enumeration, not included in this repository (Section 16), was used to recheck Lemma 3.4: all 63 traces, all $63 \cdot 63 \cdot 2$ configurations of two outside vertices, and all type subsets. The written proof above does not depend on that enumeration.

Lemma 3.5 (proved)**.** *Let G be connected, strong C_4-free and P_6-free. Then either G has a vertex of eccentricity at most 2, or $G = C_6$.*

Proof. Apply Theorem 2.5 with $k = 6$ to get a connected dominating set X. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is P_4-free. If $|X| = 1$, the vertex in X dominates G. If $|X| \geq 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since X dominates G, every vertex is within distance 2 of u. □

Lemma 3.6 (P_6-free mass; proved)**.** *If G is connected, strong C_4-free and P_6-free, then $S(G) \leq 3n$, with equality if and only if G is the Petersen graph.*

Proof. If $G = C_6$ then $S = 12 < 18$. Otherwise, by Lemma 3.5, fix u of eccentricity at most 2. Put $N_1 = N(u)$, $N_2 = V \setminus N[u]$, and let m_1 be the number of edges of $G[N_1]$. By Lemma 2.1(1) each $c \in N_2$ has exactly one neighbour in N_1, its *parent*. For $a \in N_1$ let C_a be the set of its children. An edge cd of $G[N_2]$ with distinct parents is a *cross edge*. Let E' be the set of cross edges, and let D_c be the set of cross neighbours of c.

Local values. $\lambda(u) = |N_1| - m_1$. For $a \in N_1$ with partner $\bar{a}$ (if any), no child of a is adjacent to u or $\bar{a}$, so the units of a are $\{u, \bar{a}\}$ (or $\{u\}$) together with the components of $G[C_a]$; hence $\lambda(a) = 1 + \alpha(G[C_a])$. For $c \in C_a$: c has at most one neighbour $s \in C_a$ (two would be common neighbours of a and c), the vertices of D_c are not adjacent to a, and s is adjacent to no $d \in D_c$ (else c, a, s, d is a 4-cycle). So $\lambda(c) = 1 + \alpha(G[D_c])$. Using $\sum_a |C_a| = |N_2|$ and $n = 1 + |N_1| + |N_2|$ we get

$$(2)\quad 3n - S = 3 + |N_1| + m_1 + \sum_a \big(|C_a| - \alpha(G[C_a])\big) + |N_2| - \sum_c \alpha(G[D_c]) \;\geq\; 3 + |N_1| + |N_2| - 2|E'|.$$

So it suffices to show $2|E'| \leq 3 + |N_1| + |N_2|$.

Fact 1. Let $cd \in E'$ with $c \in C_y$ and $d \in C_z$. Then $y \not\sim z$, else $ycdz$ is a 4-cycle. Let $y' \in N_1 \setminus \{y, z\}$ and $x \in C_{y'}$. Then x is adjacent to c or d. Indeed, otherwise: if $y' \not\sim y$, then $xy'uycd$ is an induced P_6; if $y' \sim y$, then $y' \not\sim z$ (since $G[N_1]$ is a matching) and $xy'uzdc$ is an induced P_6. Two children of y' adjacent to the same vertex of $\{c, d\}$ would be two common neighbours of y' and that vertex. So $|C_{y'}| \leq 2$.

Fact 2. If $E' \neq \emptyset$, at most three vertices of N_1 have children. Suppose $cd \in E'$ has parents y, z, and $y_1, y_2 \in N_1 \setminus \{y, z\}$ have children x_1, x_2. Let $H = G[\{c, d, x_1, x_2\}]$. These four vertices have distinct parents, so every edge of H is a cross edge, and by Fact 1 each of the other two vertices is adjacent to an endpoint of that edge. Hence H has no induced $K_2 \cup K_1$. So the complement of H has no induced P_3, that is, it is a disjoint union of cliques, and H is complete multipartite. A complete multipartite graph on four vertices with at least one edge and no 4-cycle subgraph is $K_{1,3}$. Let its centre be c^* and its leaves a, b, d^*, with parents a_0, b_0, d_0. Since $G[N_1]$ is a matching, two of these parents are nonadjacent, say $a_0 \not\sim b_0$. Then a_0, u, b_0, b, c^*, d^* is an induced P_6: the leaves are pairwise nonadjacent, and each child is adjacent only to its own parent in N_1. This is a contradiction. (A checker not included in this repository confirms Fact 2 by exhausting the 2^{10} adjacency patterns of the nine vertices $u, y, z, y_1, y_2, c, d, x_1, x_2$. The written argument does not use that checker.)

Counting. A child has at most one cross neighbour in each other class $C_{y'}$ (two would be common neighbours with y'), so its number of cross neighbours is at most $k - 1$, where $k \leq 3$ is the number of vertices of N_1 with children. If $E' = \emptyset$ we are done. If $k = 2$, every child has at most one cross neighbour, and $2|E'| \leq |N_2|$. Let $k = 3$, with classes C_1, C_2, C_3. If a child in C_i has two cross neighbours, in C_j and C_l, then Fact 1 applied to these two cross edges gives $|C_j|, |C_l| \leq 2$.

Distinct such children of C_i have distinct neighbours in C_j (else two common neighbours with y_i), so C_i contains at most two children with two cross neighbours, and there are at most six in total. Hence $2|E'| \le |N_2| + 6 \le |N_2| + 3 + |N_1|$, because $|N_1| \ge k = 3$.

Equality. The Petersen graph is P_6-free: if $v_0 \cdots v_5$ were an induced path, then v_0 would have a common neighbour with each of v_3, v_4, v_5 (diameter 2); these lie off the path (the only path neighbour of v_0 is v_1, which is not adjacent to them) and are distinct (girth 5), but v_0 has only two neighbours off the path. Also $\lambda \equiv 3$, so $S = 30 = 3n$. Conversely, suppose $S = 3n$. Then $G \neq C_6$; the dominating-C_6 case is excluded because there $S < 3n$ (Lemma 3.4); and $E' \neq \emptyset$, because otherwise (2) gives $3n - S \ge 3 + |N_1| + |N_2| > 0$. If $k = 2$ then $3n - S \ge 3 + |N_1| > 0$. So $k = 3$, and equality throughout forces: $|N_1| = 3$; $m_1 = 0$; no edges inside any C_a; every D_c is independent; and exactly six children have two cross neighbours, two in each class. By Fact 1 every class has at most two children, so every class has exactly two, and every child has exactly two cross neighbours. The cross edges then form a 2-regular graph on the six children with no triangle (each D_c is independent), that is, a 6-cycle. Two children of the same parent at distance 2 on this cycle would have two common neighbours, and they are not adjacent, so they are antipodal. This determines G up to isomorphism, and the Petersen graph has exactly this structure: in it, the six vertices at distance 2 from a vertex induce a 6-cycle. □

Theorem 3.7 (Theorem A; proved)**.** *Let G be connected and strong C_4-free with $n \ge 2$. If $\ell(G) \le 3$, or if $r(G) \le 2$ and $\ell(G) \le 4$, then $p(G) \ge r(G) + \ell(G)$.*

Proof. Write $r = r(G)$ and $c = \lceil \ell(G) \rceil$; we need $p \ge r + c$.

- $r = 1$: Lemma 3.1.
- $r = 2$, $c \le 2$: Lemma 3.2.
- $r = 2$, $c = 3$: if $p \le 4$, then Lemma 3.3 gives $\ell \le 2$, a contradiction. So $p \ge 5$.
- $r = 2$, $c = 4$: if $p \le 5$, then Lemma 3.6 gives $\ell \le 3$, a contradiction. So $p \ge 6$.
- $r = 3$, $c \le 2$: Theorem 2.4 gives $p \ge 5$.
- $r = 3$, $c = 3$: if $p \le 5$, then by Lemma 3.5 and $r = 3$ we have $G = C_6$, with $\ell = 2$, a contradiction. So $p \ge 6$.
- $r \ge 4$, $c \le 3$: Theorem 2.4 gives $p \ge 2r - 1 \ge r + 3$. □

Equality occurs, for instance, for C_5 ($r = 2$, $\ell = 2$, $p = 4$), the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and C_6 ($r = 3$, $\ell = 2$, $p = 5$). We do not characterise the equality cases.

4. Reduction to bounded induced paths

Theorem 4.1 (Theorem B; proved)**.** *Let G be a connected strong C_4-free graph that violates Conjecture 1.1. Then $\ell(G) > 3$. Put $K = \lceil \ell(G) \rceil - 1 \ge 3$. Then either*

(a) *$r(G) \le K$ and $p(G) \le r(G) + K \le 2K$, so G violates GEN_K; or*
(b) *$r(G) = K + 1 \ge 4$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so G violates $\mathrm{RAD}(K + 1)$.*

Proof. Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K + 1$. Then $\ell > 3$ by Theorem 3.7. A violation means $p \le r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \le p \le r + K$, so $r \le K + 1$. If $r \le K$, then $p \le r + K \le 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \le p \le 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. □

Since GEN_K ignores the radius, it is stronger than what alternative (a) needs: a counterexample in (a) also satisfies $p \le r + K$. We know of no strong C_4-free graph violating any GEN_K. Note, however, that GEN_K for large K might fail even if the conjecture holds; see Remark 9.4.

Proposition 4.2 (proved)**.** *For $k \ge 3$ let θ_k consist of two vertices x, y joined by k internally disjoint paths xa_ib_iy. Then θ_k is strong C_4-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

Proof. Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_j x a_i b_i y b_l$ (with j, i, l distinct) is induced. Conversely, let P be an induced path. If P avoids x and y, it lies inside one edge $a_i b_i$. If P contains x but not y, then x has at most two neighbours on P, each followed on P by at most its b-vertex, so $|P| \leq 5$. If P contains x and y, then every a_j on P is adjacent on P to x and every b_j to y, at most one branch lies entirely on P, and so P consists of x, y, at most one full branch and at most one further neighbour of each of x and y. Hence $|P| \leq 6$. □

So GEN_3 is sharp up to $O(1/n)$ already among graphs of radius 3, and no argument with a uniform positive margin in ℓ can prove it.

5. The case $K = 3$: a reduction for GEN_3, not a theorem

Quarantined statement 5.1 (GEN_3; not independently checkable from this archive)**.** *Every strong C_4-free graph G with no induced P_7 satisfies $S(G) \leq 3n$. Connectivity is not needed. This sentence is not a theorem of this paper: the three finite classifications below are not accompanied by their configuration lists or by a checker, and the Lean development of Section 5.3 is not in this repository.*

If Quarantined statement 5.1 were a theorem, then with Theorem 4.1 it would settle REAL-133 for every graph with $p \leq 6$, and in particular the cell $r = 3$, $3 < \ell \leq 4$. It is not a theorem, so those cells are not closed here. The argument below is a minimum-counterexample reduction. It becomes a proof only when Quarantined lemmas 5.6–5.8 are supplied with checkable enumerations.

5.1. Minimum counterexamples.

Lemma 5.2 (no rooted P_4; proved)**.** *Let F be connected and strong C_4-free with a vertex s such that no induced P_4 of F has s as an end. Then $S(F) \leq 2(|F| - 1) - m$, where m is the number of edges of $F[N(s)]$. This is* ROOT(3).

Proof. A vertex at distance 3 from s would end a geodesic, hence an induced P_4, starting at s. So $V = \{s\} \cup N_1 \cup N_2$ with N_i the i-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of y with a different parent would give the induced path $sayz$. So y has at most one neighbour in N_2, a sibling (two siblings would be common neighbours of a and y), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where C_a is the set of children of a and $\bar{a}$ is the neighbour of a in N_1, if any. Since s and $\bar{a}$ lie in one unit, $\lambda(a) \leq 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \leq (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F| - 1) - m$. □

Theorem 5.3 (proved)**.** *Let G be a counterexample to* GEN_3 *of minimum order, that is, G is connected and strong C_4-free, $p(G) \leq 6$, $S(G) \geq 3n + 1$, and n is minimum. Then:*

(a) *G is 2-connected;*
(b) *$S(G) - S(G - U) \geq 3|U| + 1$ for every nonempty proper $U \subset V(G)$;*
(c) *$2\lambda(v) - t(v) \geq 4$ for every v; in particular $\lambda(v) \geq 2$;*
(d) *every vertex with $\lambda(v) = 2$ has degree 2 and nonadjacent neighbours, and no two vertices of degree 2 are adjacent.*

Proof. (a) Since $S(G) \leq 2 < 3n$ for $n \leq 2$, we have $n \geq 3$, so it suffices to exclude cut vertices. Let v be a cut vertex. For each component C of $G - v$ let ρ_C be the largest order of an induced path in $G[C \cup \{v\}]$ with end v. Two such paths from different components meet only in v and have no edges between them, so together they form an induced path on $\rho_C + \rho_{C'} - 1 \leq 6$ vertices. So some component C_1 has $\rho_{C_1} \leq 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Both graphs are connected, smaller than G, and

P_7-free. Lemma 5.2 gives $S(G_1) \le 2(n_1-1)$, and minimality gives $S(G_2) \le 3n_2$. By Lemma 2.3, $S(G) \le 2(n_1-1)+3n_2 = 3n-(n_1-1) < 3n$, a contradiction.

(b) Every component H of $G-U$ is connected, smaller, and P_7-free, so $S(H) \le 3|H|$ (for $|H|=1$, $S(H)=0$). Summing, $S(G-U) \le 3(n-|U|)$, and $S(G) \ge 3n+1$ gives the claim.

(c) Take $U=\{v\}$ in (b) and use Lemma 2.2. Since $t(v) \ge 0$, $2\lambda(v) \ge 4$.

(d) If $\lambda(v)=2$, then (c) forces $t(v)=0$, so $d(v)=2$ and the neighbours are nonadjacent. Every vertex of degree 2 has $\lambda=2$ by (c), hence $t=0$. Let $u \sim w$ both have degree 2, with other neighbours x and y. Then $x \ne y$, since $t(u)=0$; $\{u\}$ is a unit of x and $\{w\}$ a unit of y. Lemma 2.2 with $U=\{u,w\}$ gives a loss of $2+2+1+1=6<7$, contradicting (b). □

The disconnected case follows from the connected one by summing over components (an isolated vertex has $\lambda=0$). Call a graph *admissible* if it is strong C_4-free and P_7-free, $L(v)=2\lambda(v)-t(v) \ge 4$ for every vertex v, and no two adjacent vertices have degree 2. By Theorem 5.3(b)–(d), a counterexample to GEN_3 of minimum order is admissible. In an admissible graph $\lambda \ge 2$ everywhere; $\lambda(v)=2$ forces $t(v)=0$ and $d(v)=2$, so every degree-2 vertex has nonadjacent neighbours and lies in no triangle; and every vertex of degree at least 3, a *branch vertex*, has $\lambda \ge 3$. Let B be the set of branch vertices, Z the set of degree-2 vertices (*ears*), $D=G[B]$, $d_D(v)=|N(v)\cap B|$ and $q(v)=|N(v)\cap Z|$. All triangles lie in D, so $t_D(v)=t(v)$.

Proposition 5.4 (proved from Lemma 5.5 and Quarantined lemmas 5.6–5.8)**.** *If Lemma 5.5 and Quarantined lemmas 5.6–5.8 hold, then every admissible graph satisfies $S \le 3n$, and the inequality of Quarantined statement 5.1 follows. The three finite lemmas are not independently checkable from this archive, so the inequality is not a theorem of this paper.*

Proof. Ears have nonadjacent neighbours and Z is independent, so each ear contributes $\lambda=2$ and one singleton unit to each of its two neighbours: $S(G)=S(D)+4|Z|$. Since $|Z|=\frac12\sum_{v\in B} q(v)$ and $n=|B|+|Z|$,

$$S-3n=\sum_{v\in B}\Big(d_D(v)-t_D(v)-3+\tfrac12 q(v)\Big).$$

By Lemmas 5.5, 5.7 and 5.8 every summand is at most 0: for $(d_D,t_D)=(0,0)$ we have $q\le 4$ and the summand is at most -1; for $(1,0)$ it is at most 0; for $(2,0)$ the two branch neighbours are nonadjacent, so $q\le 2$ and the summand is at most 0; for $(2,1)$, $q\le 4$ gives at most 0; for $(3,t)$, $q=0$ gives $-t\le 0$. Lemma 5.7 excludes $d_D \ge 4$. □

5.2. **Local lemmas.**

Lemma 5.5 (at most four ears; proved)**.** *In an admissible graph $q(v)\le 4$ for every v.*

Proof. Let u_i be ears at v with other neighbours a_i. The a_i are distinct (else vu_iau_j is a C_4), nonadjacent to v (ears lie in no triangle) and of degree at least 3 (no adjacent degree-2 pair). Put $E_i=N(a_i)\setminus\{u_i\}$; then $|E_i|\ge 2$, and $E_i\cap N[v]=\emptyset$, since a second common neighbour of v and a_i would give a C_4.

Claim. If $a_i \not\sim a_j$, then $E_i=\{w,p\}$, $E_j=\{w,q\}$ with $pq\in E$, $w\not\sim p$ and $w\not\sim q$; hence a_i,a_j have degree 3 and lie in no triangle. Indeed $|E_i\cap E_j|\le 1$; for $p\in E_i\setminus E_j$ and $q\in E_j\setminus E_i$ with $p\not\sim q$ the path $pa_iu_ivu_ja_jq$ is induced (all chords are excluded by $d(u_i)=d(u_j)=2$, $E_i\cap N[v]=E_j\cap N[v]=\emptyset$, $a_i\not\sim a_j$ and the choice of p,q). So the private parts are complete to each other. Both are nonempty, and if one had two vertices p,p', then a_ipqp' would be a C_4; so both have size one and $|E_i\cap E_j|=1$. An edge wp would give the C_4 wa_jqp.

No three pairwise nonadjacent a_i. Otherwise the sets E_i pairwise meet in exactly one element. If the three intersections are the three pairs of a 3-set $\{\alpha,\beta,\gamma\}$, the claim forces the triangle $\alpha\beta\gamma$, and $a_1\sim\alpha,\beta$ gives the C_4 $a_1\alpha\gamma\beta$. Otherwise $E_i=\{w,p_i\}$ with $p_1p_2p_3$ a triangle and $w\not\sim p_i$, and $vu_1a_1wa_2p_2p_3$ is an induced P_7.

Five ears. Then $F = G[\{a_1, \dots, a_5\}]$ has $\alpha(F) \le 2$. If F has a vertex a adjacent to the other four, these induce a matching (Lemma 2.1), and since $\alpha(F) \le 2$ it is a perfect matching $\{b_1b_2, c_1c_2\}$; $b_1 \not\sim c_1$, so by the claim b_1 lies in no triangle, but ab_1b_2 is one. Otherwise, by the claim every vertex of F that has a non-neighbour in F has degree 3 in G and lies in no triangle; so F is triangle-free with maximum degree at most 2, C_4-free and has $\alpha(F) \le 2$, hence $F = C_5 = a_0 \cdots a_4$, and $u_0vu_1a_1a_2a_3a_4$ is an induced P_7. □

Attachment certificates. Let H be an induced subgraph of an admissible graph G, let $R \subseteq V(H)$ be a set of vertices known to have all their G-neighbours in H, and $K \subseteq V(H)$ a set of vertices known to be branch vertices. A *kept trace* is a nonempty $A \subseteq V(H) \setminus R$ such that H plus one new vertex with neighbourhood A is strong C_4-free and P_7-free. Call $x \in V(H)$ *raisable* if some kept trace contains x and misses $N_H(x)$. If x is not raisable, every neighbour y of x outside H has a kept trace $N(y) \cap V(H)$ (since $H + y$ is induced in G) containing x and some old neighbour z of x; so xyz is a new triangle, y joins the previously singleton unit of z, and $\lambda_G(x) = \lambda_H(x)$ and $L_G(x) \le L_H(x) -$ (number of new neighbours). Hence H cannot occur in an admissible graph if it has (i) a nonraisable x with $L_H(x) < 4$, or (ii) a nonraisable $x \in K$ with $\lambda_H(x) < 3$, or (iii) an edge xy with $d_H(x) = d_H(y) = 2$, both ends nonraisable and $L_H(x) = L_H(y) = 4$ (they would remain adjacent degree-2 vertices of G). No bound on $|V(G)|$ is involved.

Quarantined lemma 5.6 (two separated forks; not independently checkable from this archive)**.** *An admissible graph contains no induced 7-vertex tree T with edges $va, vb, aa_1, aa_2, bb_1, bb_2$.*

Proof outline. Since $\lambda(a_i) \ge 2$ there is $A_i \in N(a_i) \setminus N[a]$, and similarly $B_j \in N(b_j) \setminus N[b]$. These lie outside T, are nonadjacent to v (a C_4 with a or b) and are pairwise distinct: on the same side by C_4-freeness, and $A_i = B_j$ by the induced path $a_{3-i}aa_iA_ib_jbb_{3-j}$. Call A_i *targeted* if $A_i \sim b$ (and B_j if $B_j \sim a$). A case analysis with explicit induced P_7's shows that all four witnesses are targeted. Then a, b are nonadjacent hubs with common neighbour v and four internally disjoint paths ax_iy_ib ($X = \{a_1, a_2, B_1, B_2\} \subseteq N(a)$, $Y = \{A_1, A_2, b_1, b_2\} \subseteq N(b)$). On these 11 vertices the only undetermined pairs are matching edges inside X and inside Y. Of the $10 \times 10 = 100$ configurations, 79 are strong C_4-free, 43 are also P_7-free, and each of the 43 is excluded by an attachment certificate with $R = K = \emptyset$ (2047 candidate traces each, of which 5 to 13 are kept). □

Quarantined lemma 5.7 (at most three branch neighbours; not independently checkable from this archive)**.** *In an admissible graph $d_D(v) \le 3$ for every v.*

Proof outline. Let $u_1, \dots, u_4$ be branch neighbours of v. As $\lambda(u_i) \ge 3$, pick x_i, y_i in two distinct units of u_i other than the unit of v; they are nonadjacent and lie outside $N[v]$. These child classes are disjoint, no child is adjacent to another selected parent, and between two classes the edges form a matching. The parents induce a matching. Adjacent parents have anticomplete classes (C_4), and nonadjacent parents have at least one edge between their classes, since otherwise v, u_i, u_j and the four children induce the tree of Lemma 5.6. So each nonadjacent pair of parents has 6 options. Of the $6^6 = 46656$, $6^5 = 7776$ and $6^4 = 1296$ configurations (no, one, two parent edges), 24688, 6288 and 1048 are strong C_4-free and none is P_7-free. □

Quarantined lemma 5.8 (mixed neighbourhoods; not independently checkable from this archive)**.** *In an admissible graph:* (a) *$d_D(v) = 3$ implies $q(v) = 0$;* (b) *if $d_D(v) = 2$ and the two branch neighbours are nonadjacent, then $q(v) \le 2$.*

Proof outline. Select three branch neighbours and one ear for (a), and two nonadjacent branch neighbours and three ears for (b). Branch parents get two children as in Lemma 5.7; an ear parent has a unique child (its other neighbour), a branch vertex nonadjacent to v and to the other parents and distinct from the other children. Ear–branch class pairs have 3 options, ear–ear pairs 2, nonadjacent branch–branch pairs 6 (Lemma 5.6), and adjacent branch parents are

anticomplete. Use $R =$ the ear parents and $K = \{v\} \cup$ branch parents $\cup$ ear children. For (a) with independent parents there are 5832 configurations, 4320 strong C_4-free and 32 P_7-free, all excluded by attachment certificates; with one parent edge, 972, 876 and 0. For (b), 34992, 17088 and 18, all 18 excluded. □

The finite parts of Lemmas 5.6–5.8 were reproduced by three independent implementations (Section 15). As a sanity check (not used in the proof), among all connected strong C_4-free P_7-free graphs with $n \leq 14$ the only admissible graphs are four graphs on 8, 9, 10 and 11 vertices, one of them the Petersen graph. Earlier versions of this paper reduced GEN_3 to a search over "skeletons" of branch vertices and verified it for $n \leq 20$; that material is superseded by Theorem 5.1 and has been removed.

5.3. **Lean formalisation.** The development described in this subsection is not a file in this repository. It is not a check available to a reader of the archive, and it does not place GEN_3 on the claim ladder. What follows is a description of a formalisation that was carried out elsewhere. The statement that was formalised is the inequality of Quarantined statement 5.1, in Lean 4 (v4.35.0) with Mathlib [8, 7]. The main declaration reads

```
theorem gen3 (hC : StrongC4Free G) (hP : P7Free G) : S G ≤ 3 * Fintype.card V
```

for an arbitrary finite simple graph `G` on a vertex type `V`; here `StrongC4Free` says that two distinct vertices have at most one common neighbour, `P7Free` says that no injective map $\mathrm{Fin}\,7 \to V$ is an induced path, and `S G` is the sum over v of the independence number of the induced subgraph on $N(v)$ (Mathlib's `indepNum`). The formalisation follows the proof above with one simplification: a single generic configuration checker, with a machine-checked soundness theorem, verifies case-split trees whose leaves are C_4 witnesses, induced-P_7 witnesses, degree witnesses, contradictions, or iterated attachment extensions. With it, Lemmas 5.5, 5.7, 5.8(a) and 5.8(b) become small hand-written setups plus one certificate each (1008, 14969, 6419 and 3828 nodes); Lemma 5.6 is not needed. All finite checks run in the kernel by `decide +kernel` (no `native_decide`). The axioms of `gen3` are `propext`, `Classical.choice` and `Quot.sound`; the certificate checks use only `propext`. A clean build of the development (with Mathlib cached) takes about three minutes.

6. The case $K = 4$: GEN_4, not a theorem

Quarantined statement 6.1 (GEN_4; not independently checkable from this archive)**.** *Every connected strong C_4-free graph with no induced P_9 satisfies $S(G) \leq 4n$. The extension DAG named in the outline is not in this repository, so the sentence is not a theorem of this paper.*

Proof outline. (i) Let G be a counterexample of minimum order, so $p(G) \leq 8$ and $S(G) \geq 4n+1$. Every component of $G - U$ is smaller, so $D_G(U) \geq 4|U| + 1$ for every nonempty proper U; in particular $L(v) \geq 5$ and $\lambda(v) \geq 3$ for every v.

(ii) *Attachment lemma.* If H is an induced subgraph of G and $x \in V(H)$ has $L_H(x) < 5 \leq L_G(x)$, then some $y \notin V(H)$ adjacent to x has $N(y) \cap N_H(x) = \emptyset$. Otherwise every outside neighbour of x joins a distinct old singleton unit of x, so $\lambda_G(x) = \lambda_H(x)$ and $L_G(x) \leq L_H(x)$. The trace $A = N(y) \cap V(H)$ contains x, misses $N_H(x)$, and $H + y$ is strong C_4-free and P_9-free.

(iii) *Finite lemma.* No strong C_4-free P_9-free graph with $L \geq 5$ everywhere contains an induced $K_{1,5}$. The certificate is a finite rooted directed acyclic graph of records. The root is $K_{1,5}$. Each record H names a vertex x with $L_H(x) < 5$ and lists, for *every* trace A as in (ii), a child record isomorphic to $H + y$, with an explicit isomorphism. There are 3796 records (2287 terminal, that is, without any admissible trace), 6679 arcs and 1,929,045 examined C_4-safe traces; the largest record has 27 vertices. If G contained $K_{1,5}$, following (ii) through the certificate, carrying an induced embedding into G, would reach a terminal record, a contradiction. Every arc raises the order by one, so the walk terminates; no bound on $|G|$ is assumed.

(iv) A vertex with $\lambda \geq 5$ yields an induced $K_{1,5}$ (one vertex per unit). So $3 \leq \lambda \leq 4$ everywhere and $S - 4n = -\#\{v : \lambda(v) = 3\} \leq 0$, a contradiction. □

The bound $\lambda \leq 4$ in (iv) is attained: C_8 with four vertices $a_i \sim c_i, c_{i+4}$ and a root adjacent to all a_i has 13 vertices, $p = 7$ and $\lambda = 4$ at the root. If Quarantined statement 6.1 were a theorem, then with Theorem 4.1 it would show that $\ell > 4$ implies $p \geq 9$. That implication is not claimed.

7. The radius-extremal side: RAD(r) for $r \leq 7$

7.1. **Radius four.** A finite classification settles the smallest case RAD(4).

Quarantined lemma 7.1 (dominating 8-cycle; not independently checkable from this archive)**.** *Let G be connected, strong C_4-free and P_8-free, with a dominating induced 8-cycle. If $r(G) \geq 4$, then $G = C_8$.*

Proof. Let C be the cycle and $W = V(G) \setminus V(C)$. Of the 255 nonempty traces on C, exactly 32 give a strong C_4-free, P_8-free graph $C + w$, and each of them has at least two vertices. So two vertices of W with the same trace would have two common neighbours, and the vertices of W have distinct types. List them in increasing order of type; every prefix induces a strong C_4-free, P_8-free subgraph, since both properties are hereditary. Adding vertices of increasing type, with every adjacency to earlier vertices and pruning at a 4-cycle or an induced P_8, produces 790 such configurations (73 isomorphism classes), with at most 13 vertices; the bound 13 is an outcome of the search, not an input. Among them only C_8 itself has radius at least 4. □

Quarantined statement 7.2 (radius four; not independently checkable from this archive)**.** *Let G be connected and strong C_4-free with $r(G) = 4$ and $p(G) = 7$. The trace argument below shows $S(G) \leq 2n$ once the dominating core is a 6-cycle. The remaining case, an 8-cycle core, uses Quarantined lemma 7.1, whose list of 790 configurations is not in this archive. The inequality $S(G) \leq 2n$ is therefore not a theorem of this paper.*

Proof. By Theorem 2.5 with $k = 8$, G has a connected dominating set X with $G[X]$ P_6-free or $G[X] \cong C_8$. In the second case Lemma 7.1 gives $G = C_8$ and $\ell = 2$. In the first case, by Lemma 3.5 applied to $G[X]$, either $G[X]$ has a vertex of eccentricity at most 2 in $G[X]$, which then has eccentricity at most 3 in G since X dominates G, contradicting $r(G) = 4$; or $G[X] = C_6$. So let $C = v_0 \cdots v_5$ (indices modulo 6) be a dominating induced 6-cycle and $W = V(G) \setminus V(C)$.

Traces. As in the proof of Lemma 3.4, strong C_4-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some w has trace $\{v_i, v_{i+3}\}$, then v_i is within distance 2 of every vertex of C, hence within distance 3 of every vertex of G, a contradiction. So there are no antipodal traces. Let C_i be the set of vertices with trace $\{v_i\}$.

Witnesses. Fix i. Since $r(G) = 4$, some vertex is at distance at least 4 from v_{i+3}. Every vertex of C is within distance 3 of v_{i+3}, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from v_{i+3}, which is at most 3 unless the trace of w is $\{v_i\}$. So there is $w_i \in C_i$ at distance 4 from v_{i+3}. Every neighbour $z \in W$ of w_i lies in C_i: if z had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \leq 3$; if z had the neighbour v_{i-1} or v_{i+1}, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.

Edges in W. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ is an induced P_8.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced P_8. By symmetry, vertices of C_i and C_j with $i \neq j$ are never adjacent.

- x with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices v_0, v_1, x, y span a 4-cycle; for $j = 5$ (resp. $j = 2$) the cycle x, v_0, v_5, y (resp. x, v_1, v_2, y) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced P_8, and $j = 4$ is symmetric.
- x, y with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm 1$ and $j = \pm 2$ give 4-cycles as above; $j = 3$ gives the induced P_8 $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

In each displayed path the non-edges follow from the traces and from $N(w_i) \subseteq \{v_i\} \cup C_i$. So every edge inside W joins two vertices of the same class C_i, and these edges form a matching (two neighbours in C_i of one vertex would be two common neighbours with v_i).

Mass. Let $s = \sum_i |C_i|$, let t be the number of vertices with a consecutive-pair trace, and let m be the number of edges inside W. Then $n = 6 + s + t$, $e(G) = 6 + s + 2t + m$, and $T(G) = t + m$. So $S = 2e - 3T = 12 + 2s + t - m = 2n - t - m \leq 2n$. □

Corollary 7.3 (proved)**.** *Let G be connected and strong C_4-free. If $r(G) \geq 5$ and $\ell(G) \leq 4$, then $p(G) \geq r(G) + \ell(G)$.*

Proof. Theorem 2.4 gives $p(G) \geq 2r(G) - 1 \geq r(G) + 4 \geq r(G) + \lceil \ell(G) \rceil$. □

Quarantined statement 7.4 (depends on the radius-four bound; not a theorem)**.** *If Quarantined statement 7.2 were a theorem, then REAL-133 would hold for every connected strong C_4-free graph with $r(G) = 4$ and $\ell(G) \leq 4$, and* RAD(4) *would hold. The radius-four bound is not independently checkable from this archive, so neither conclusion is claimed. The case $r \geq 5$ and $\ell \leq 4$ is Corollary 7.3 and does not use that bound.*

Proof. The following calculation is conditional on Quarantined statement 7.2. Let $r = 4$. By Theorem 2.4, $p \geq 7$. If $p = 7$, the radius-four bound would give $\ell \leq 2$. If $p \geq 8$, then $p \geq r + 4 \geq r + \lceil \ell \rceil$ with no appeal to that bound. □

7.2. **Radius five, six and seven.** The next three statements follow the pattern of Quarantined statement 7.2. Theorem 2.5 produces a dominating core, and the vertices outside it are controlled by trace classifications. The outlines below are not proofs. The trace lists and recomputations were written as supplementary reports; those files are not in this repository (Section 16), so the three inequalities are not independently checkable from this archive and are not theorems.

Quarantined statement 7.5 (RAD(5); not independently checkable from this archive)**.** *If G is connected, strong C_4-free, of radius 5 and P_{10}-free, then $e(G) = n + T(G)$, so $S(G) = 2n - T(G) \leq 2n$ and $\ell \leq 2$.*

Proof outline. Theorem 2.5 with $k = 10$ gives a connected dominating H that is C_{10} or P_8-free. In the latter case $r(H) = 4$, $p(H) = 7$, and the proof of Theorem 7.2 shows that $H = C_8$ or H has a dominating C_6 with singleton or consecutive-pair traces and same-class matchings. (i) A dominating C_{10}: an outside vertex has neither a singleton nor a consecutive-pair trace (each gives a P_{10}), so its trace contains c_0, c_d with $3 \leq d \leq 5$; the shortcut gives $\mathrm{ecc}(c_0) \leq 4$. So $G = C_{10}$. (ii) A dominating C_8: traces are singletons or consecutive pairs; witnesses at distance 5 from the antipodal cycle vertex have singleton traces and are independent, which gives an induced C_8 with a pendant vertex at each cycle vertex. The classification has 24 trace types meeting C_8, and adjacent outside vertices lie in a common class; counting gives $e = n + T$. (iii) A dominating C_6: witnesses give an induced C_6 with a pendant P_2 at each vertex; the classification has 31 single and 31 adjacent-pair trace types, every vertex with empty trace has exactly one parent, and edges between such vertices join vertices with the same parent; again $e = n + T$. □

The bound is attained by C_8 with a pendant vertex at each cycle vertex. The strong identity $e = n + T$ fails for every $r \geq 6$ (Example 11.1), but the weaker bounds needed for RAD(r) survive.

Quarantined statement 7.6 (RAD(6); not independently checkable from this archive)**.** *If G is connected and strong C_4-free with $r(G) = 6$ and $p(G) = 11$, then $S(G) \le 5n - 36$.*

Proof outline. Theorem 2.5 with $k = 12$ gives $H = C_{12}$ (then $G = C_{12}$ by the dominating-cycle shortcut argument) or a P_{10}-free H with $r(H) = 5$, $p(H) = 9$, which has the structure from Theorem 7.5. Witnesses at distance 6 extend this to an induced *flower* $F_{L,t}$: a cycle C_L with a pendant path on t vertices at each cycle vertex, $(L, t) \in \{(6, 3), (8, 2), (10, 1)\}$. Put $A = N(C) \setminus C$ and $R = V \setminus (C \cup A)$. Complete finite classifications give the single traces (43, 41, 31 types) and adjacent pairs (55, 41, 11), and show that there is no induced path $a\,x\,y\,z$ with $a \in A$ and $x, y, z \in R$, and that every vertex of R has at most two neighbours in A. A written mass lemma bounds each component J of $G[R]$: if every vertex has at most d parents and no parent starts an induced P_4 into J, then $S(J) + 2e(A, J) - 3\tau \le (2d + 1)|J|$, where $\tau = T_{A,J,J}$ is the number of triangles with one vertex in A and two in J. With $S(G[C \cup A]) \le 2|C \cup A|$ and $d = 2$ this gives $S \le 5n - 36$. □

Quarantined statement 7.7 (RAD(7); not independently checkable from this archive)**.** *If G is connected and strong C_4-free with $r(G) = 7$ and $p(G) = 13$, then $S(G) \le 5n - 42$, with equality if and only if $G = C_{14}$.*

Proof outline. The structure from Theorem 7.6 propagates one level (the *lifting lemma*) and gives a flower $F_{L,t}$ with $L + 2t = 14$, $(L, t) \in \{(12, 1), (10, 2), (8, 3), (6, 4)\}$, or the cycle C_{14}. The single traces (37, 51, 57, 55 types) and adjacent pairs (13, 51, 73, 91) are classified; rooted paths of the form $A\,R^k$ and vertices of R with three parents in A are excluded. For $(6, 4)$ three further facts hold: a two-parent vertex has empty trace and nonadjacent parents with supports c_i, c_{i+1}; it starts no induced P_4 in $G[R]$; and two such vertices are never at distance 2. Hence the two-parent vertices of each component of $G[R]$ form a clique of order at most 3, so its mass is at most $5|J|$. Tip components save $L(t - 1)$, giving $S \le 5n - 54$ for flowers, and the cycle gives $5n - 42$. □

These structural statements are the first instances of a uniform structure theorem, Struct(r), stated precisely in Appendix A (Definition A.1 and Theorem A.2): a connected strong C_4-free graph of radius $r \ge 5$ with $p = 2r-1$ is C_{2r} or contains an *admissible* induced flower $F_{2s,t}$ ($s+t = r$, $s \ge 3$, $t \ge 1$; admissibility includes the ambient distance inequalities $d_G(c_{i+s}, z_i) \ge r$ for the arm tips z_i), and the vertices outside the flower satisfy explicit trace, matching, rooted-path and two-parent restrictions. The base case Struct(5) is supplied by the proof of Theorem 7.5; the lifting Struct(r) ⇒ Struct($r + 1$) has a uniform written proof. The numerical bounds of Theorems 7.5–7.7 alone would not suffice for this: the lifting uses the structural conclusions, in particular the distance inequalities, which cannot be imposed on arbitrary induced templates. What Struct(r) does not provide is control of the mass of the components of $G[R]$ hanging from the flower; that is the role of the rooted bounds of the next section.

8. Rooted mass bounds and sharp RAD(r) for $r \le 10$

Recall that $p_s(F)$ is the largest order of an induced path of F starting at s, $D = d_F(s)$ and $m = e(F[N(s)])$.

Quarantined statement 8.1 (ROOT(q) for $4 \le q \le 8$; not independently checkable from this archive)**.** *Let F be connected and strong C_4-free, $s \in V(F)$ and $3 \le q \le 8$. The inequality*

$$S(F) \le (q - 1)(|F| - 1) - (q - 3)D - m$$

is proved for $q = 3$, by Lemma 5.2. For $q = 4$ and $q = 5$ the written proofs are the supplementary reports `lanes/astra_r3b/rad K2/scratch/RAD6_ROOT_REPORT.md` *and* `lanes/astra_r4/rad7/scratch/RAD7_ROOT5_REPORT.md`*. Those reports are not in this repository. For $6 \le q \le 8$ the argument uses enumerations that are not in this repository. The inequality for*

$4 \le q \le 8$ is not a theorem of this paper. Where a leading coefficient $q-1$ is fixed, the corrections $q-3$ and 1 are recorded as optimal; that optimality is part of the same missing arguments.

q	bound	status	main finite ingredients
3	$2(n-1)-m$	proved in the text	none
4	$3(n-1)-D-m$	supplement absent	not checkable here
5	$4(n-1)-2D-m$	supplement absent	charging argument not in the archive
6	$5(n-1)-3D-m$	payload absent	counts named in an absent certificate
7	$6(n-1)-4D-m$	payload absent	counts named in an absent certificate
8	$7(n-1)-5D-m$	payload absent	counts named in an absent certificate

TABLE 2. ROOT(q) for $q \le 8$ ($n = |F|$).

Proof outline. Component form. A *parent system* is a connected graph J with $h = |J|$, a nonempty independent set B of parents, each with a neighbour in J, and edges from B to J such that each vertex of J has at most one parent and the whole graph is strong C_4-free. Its boundary mass is $M_B(J) = S(J) + 2e(B,J) - 3\tau$, where $\tau = T_{B,J,J}$ counts triangles with one vertex in B and two in J. (Edges between parents enter neither $M_B(J)$ nor the hypothesis below, so they may be present or deleted.) Statement C_q says: if, for every $u \in B$, every induced path of $J+u$ starting at u has at most $q-1$ vertices, then $M_B(J) \le (q-1)h$. Closed-neighbourhood amalgamation of many copies of F at the root (which preserves p_s and C_4-freeness), together with auxiliary roots and summation over components, shows that C_q, ROOT(q) and the corrected form with the terms $-(q-3)D-m$ are equivalent. The same amalgamation applied to universal-root matching graphs proves optimality of the corrections.

Minimum counterexamples. In a counterexample to C_q with h minimum, every nonempty $U \subseteq V(J)$ has deletion loss $\Delta_B(U) \ge (q-1)|U|+1$ (components of $J-U$ that lose all parents are bounded by ROOT($q-2$) after a rooted prefix). Averaging the corrected ROOT($q-1$) over the parents proves C_q whenever there are at most $q-2$ parents, so a counterexample has at least $q-1$ parents. For $q=8$, each parent has at least two children, and $M_B(J) > 7h$ and $e(B,J) \le h$ give $S(J) > 5h$, so GEN_4 (Theorem 6.1) forces an induced P_9 in J. The proof then excludes, by complete finite enumeration of the possible traces of additional parents, every way in which the parents can meet such a path. For $q=8$, all parents lie within distance 2 of every induced P_9, and the configurations with $6, 5, \dots, 0$ direct parents on a P_9 are excluded in turn. □

Quarantined statement 8.2 (sharp RAD(r) for $5 \le r \le 10$; not independently checkable from this archive)**.** *Let $5 \le r \le 10$ and let G be connected and strong C_4-free with $r(G) = r$ and $p(G) = 2r-1$. The calculation below yields*

$$S(G) \le (r-1)n - 2r(r-3),$$

with equality only for C_{2r}, and the stricter non-cycle deficits $\delta_r = 32, 60, 96, 126, 156, 186$ for $r = 5, \dots, 10$, if Struct(r) *and* ROOT(q) *for $4 \le q \le 8$ are granted. Both inputs are quarantined. The inequality is not a theorem of this paper.*

Proof. By Theorem A.2, either $G = C_{2r}$, which has $n = 2r$, $S = 4r = (r-1)n - 2r(r-3)$, or G contains an admissible flower $F_{L,t}$ with $L = 2s$, $s+t = r$, $s \ge 3$, $t \ge 1$, satisfying (S1)–(S5) of Definition A.1. Let C, A, R be as there and $n_0 = |C \cup A|$. The first arm vertices lie in A, so $n_0 \ge 2L$, and the flower has $L(t+1)$ vertices, so $n \ge L(t+1)$.

Decomposition. No vertex of R is adjacent to C, and two A-neighbours of an R-vertex are nonadjacent by (S5). Hence every triangle meeting R has one vertex in A and two in R, and

$$S(G) = S(G[C \cup A]) + \sum_J M_A(J),$$

the sum over the components J of $G[R]$, with $M_A(J)$ the boundary mass of Section 8 for the actual parents of J. By (S1)–(S2) the A-vertices have singleton or consecutive-pair traces and the edges of $G[A]$ join equal singleton traces and form a matching; counting edges and triangles gives $S(G[C \cup A]) \le 2n_0$.

Case $t = 1$. Then $R = \varnothing$ by (S4), $S \le 2n_0 = 2n$, and $n \ge 2L = 4(r-1)$. So $(r-1)n - S \ge (r-3)n \ge 4(r-1)(r-3)$.

Case $t \ge 2$. Put $\beta = t$. Since $t \le r - 3 \le 7$, Theorem 8.1 gives ROOT$(t+1)$, which in component form is $M_B(J) \le t|J|$ whenever every parent $u \in B$ has $p_u(J+u) \le t$ and every vertex of J has at most one parent in B. By (S3) every parent of J satisfies the rooted condition; by (S5) the parents split into two classes A_0, A_1 such that each vertex of J has at most one parent in each. Applying the component bound to the class that meets J (either one if both do) and charging the other class at most 2 per vertex of J gives $M_A(J) \le (\beta+2)|J|$. For each i, the arm vertices $v_{i,2}, \dots, v_{i,t}$ lie in one component J_i of $G[R]$, these L components are distinct, and all parents of J_i have the singleton trace $\{c_i\}$ (this follows from the distance inequality for z_i; Appendix A). So J_i has unique parents and $M_A(J_i) \le \beta|J_i|$. Moreover the tip z_i is a leaf or a degree-two triangle vertex of G: along a geodesic $x_0, \dots, x_{t-1} = z_i$ in $J_i + x_0$ from a parent x_0, an extra neighbour w of z_i must meet the geodesic (else the rooted path extends), a contact before x_{t-3} shortens the distance, a contact at x_{t-3} gives a 4-cycle, so $w \sim x_{t-2}$, and two such w give a 4-cycle. Deleting z_i therefore costs at most 2, and $M_A(J_i) \le \beta(|J_i| - 1) + 2$. Summing, with $\sum_i |J_i| \ge L(t-1)$,

$$S \le 2n_0 + (\beta+2)(n - n_0) - 2L(t-1) - L(\beta-2) \le (\beta+2)n - L(3\beta + 2t - 4).$$

With $\beta = t$ and $n \ge L(t+1)$ the deficit from $(r-1)n$ is at least $(s-3)n + 2s(5t-4) \ge 2s\big[(s+2)t + s - 7\big]$.

Comparison. For $5 \le r \le 10$ the minimum of these non-cycle deficits over $3 \le s \le r-1$ is δ_r as stated (attained at $s = r-1$, $t = 1$ for $r = 5, 6$, and at $s = 3$ for $7 \le r \le 10$), and $\delta_r > 2r(r-3) = 20, 36, 56, 80, 108, 140$. So every non-cycle satisfies the bound strictly and equality holds only for C_{2r}. □

For $r = 8$ the strict form of ROOT(6) improves the non-cycle constant to 132 (supplement, `lanes/astra_r7/radpush`). For $r = 5, 6, 7$ the inequality (but not the equality statement) also follows from Theorems 7.5–7.7 and $n \ge 2r$. For $r \ge 11$ the same computation needs ROOT(q) for $q \le r - 2$, and gives the conditional bound below.

Quarantined statement 8.3 (deduction only; the range $\ell \le 5$ is not claimed)**.** *Suppose Quarantined statements 5.1, 6.1, 7.2 and 7.5 were theorems. Then REAL-133 would hold for every connected graph with $\ell(G) \le 5$, and, granting also Quarantined statements 7.4 and 8.2,* RAD(r) *would hold for $4 \le r \le 10$. Those inputs are not independently checkable from this archive. REAL-133 for $3 < \ell \le 5$ is not a theorem of this paper, and REAL-133 remains open.*

Proof. The following calculation is conditional on the quarantined inputs just named. It does not establish its conclusion. Theorem 3.7 covers $\ell \le 3$. For $3 < \ell \le 5$, $K = \lceil \ell \rceil - 1 \in \{3, 4\}$. Alternative (a) of Theorem 4.1 would be excluded by GEN_3 and GEN_4. Alternative (b) has $r = K + 1 \in \{4, 5\}$ and $p = 2r - 1$, and the radius-four and radius-five bounds would give $\ell \le 2 < r - 1$. The last clause of the statement would follow in the same conditional sense, since $(r-1)n - 2r(r-3) < (r-1)n$. □

Beyond $q = 8$. The following sentences are quarantined. An *outer component* is a component of $F - N[s]$, and its parents are its neighbours in $N(s)$. None of them is a theorem of this paper.

- *Corrected* ROOT(9) *with few parents* (payload absent; it uses ROOT(8)): if $p_s(F) \le 9$ and every outer component has at most seven parents, then the recorded bound is $S(F) \le 8(n-1) - 6D - m$.
- *Ordinary* ROOT(9) *for small order* (payload absent): if $p_s(F) \le 9$ and $|F| \le 190$, then the recorded bound is $S(F) \le 8(|F|-1)$. This is the weaker inequality without the terms $-6D - m$. The amalgamation that adds those terms uses arbitrarily many copies of F and does not apply to a statement restricted to bounded order.
- *Finite local criterion* (the supporting write-up is not in this repository): for $q \ge 4$, ROOT$(q-1)$ together with $\mathcal{L}_q(b)$ (Definition B.2) would imply ROOT(q). Quarantined lemma B.1 and Quarantined statement B.3 record that implication. For $q = 9$, all 350 seeds of $\mathcal{L}_9$ are open under the depth-zero rules. ROOT(9) is not a theorem.
- *Conditional comparison* (arithmetic written; hypotheses quarantined): if Struct(r) holds and ROOT(q) holds for $3 \le q \le r-2$, then the comparison in the proof of Quarantined statement 8.2 gives $S \le (r-1)n - \min\{2r(r-3),\ 30r-114\}$ for every connected strong C_4-free graph of radius $r \ge 5$ and path order $2r-1$. For $t \ge 2$ the minimum of $2s\lfloor(s+2)t+s-7\rfloor$ over $3 \le s \le r-2$ is attained at an endpoint, with values $30r - 114$ and $6(r-2)(r-3)$, and the case $t = 1$ gives $4(r-1)(r-3)$. The hypotheses are not theorems, so neither is the bound.

9. Restricted classes: all indices

9.1. **Blocks that are bipartite or cycles.** Let $\mathcal{B}$ be the class of connected strong C_4-free graphs in which every block is bipartite or a cycle (edges are blocks; cycles may be odd). It contains every connected strong C_4-free bipartite graph and every connected strong C_4-free cactus, but of course not graphs containing a 4-cycle: the restriction matters, since $K_{2,3}$ rooted in its two-vertex part has $p_s = 3$ and $S = 12 > 2(n-1) - m = 8$. A rooted graph (G, s) with $p_s(G) = q$ and $\mathrm{ecc}(s) = q - 1$ is called *terminal*; for it put $R = n - q + 1 - D \ge 0$, the number of vertices outside $N[s]$ and outside a root-to-farthest-vertex geodesic. Set $c_2 = 0$, $c_3 = 2$, $c_4 = 3$ and $c_q = 2\lfloor(q-1)/2\rfloor$ for $q \ge 5$.

Theorem 9.1 ((1) and (2) proved; (3) not a theorem)**.** (1) *Every terminal instance in $\mathcal{B}$ satisfies $S(G) \le 2(D + q - 2) - m + c_q R$.*

(2) *Every $F \in \mathcal{B}$ with $p_s(F) \le q$ satisfies $S(F) \le c_q(n-1) - (c_q - 2)D - m$; in particular* ROOT$(q)$ *holds in $\mathcal{B}$ for every $q \ge 3$. If every block is bipartite or a triangle, one may take $c_q = 2\lfloor(q-1)/2\rfloor$ for all $q \ge 3$.*

(3) (*not a theorem; uses* Struct(r)*, which is not independently checkable from this archive*) *For every $r \ge 5$, every $G \in \mathcal{B}$ with $r(G) = r$ and $p(G) = 2r-1$ satisfies $S \le (r-1)n - 2r(r-3)$, with equality exactly for C_{2r}; if G is not a cycle, $S \le (r-1)n - 4(r-1)(r-3)$, with equality exactly for C_{2r-2} with one pendant vertex at each cycle vertex. Part (3) is not claimed.*

Proof. Two terminal facts. An endpoint x of a longest rooted induced path is not a cut vertex: a neighbour of x in another component of $G - x$ would extend the path. If z is at distance $q-1$ from s and $v_0 = s, \dots, v_{q-1} = z$ is a geodesic, any neighbour $y \ne v_{q-2}$ of z must meet an earlier path vertex (else the rooted path extends); meeting v_i with $i \le q-4$ shortens $d(s, z)$, and meeting v_{q-3} gives a C_4; so $y \sim v_{q-2}$, and two such y would give a C_4. Hence $L(z) = 2d(z) - 3t(z) \le 2$. Moreover, if $x \ne s, z$, $G - x$ is connected and z survives, then $q - 1 \le d_{G-x}(s, z) \le p_s(G - x) - 1 \le q - 1$, so the instance stays terminal with the same q.

Endpoint losses. Let x be the endpoint of a longest rooted path of order q. It lies in a single block. If that block is a triangle the loss $L(x)$ is 1, if an edge 2, if a cycle of length at least 5 it is 4. If it is bipartite, let ℓ be the order of the final path segment inside it. Block-tree separation shows that the neighbours of x in this block can only be blocked by earlier path vertices in the same block; every neighbour other than the predecessor has a blocker in the colour class of x, the vertex two positions earlier cannot block (a C_4), and each position blocks at most one neighbour. So $d(x) \le \lfloor(\ell-1)/2\rfloor \le \lfloor(q-1)/2\rfloor$, there are no triangles at x, and $L(x) \le c_q$ for $q \ge 5$.

Bases. For $q \le 4$ a nontrivial bipartite block would yield, by the same endpoint count, a rooted induced path of order at least 5, so these instances are cacti. For a C_4-free cactus with c_3 triangle blocks and c_L longer cycle blocks, $S = 2(n-1) + 2c_L - c_3$ and $m \le c_3$. For $q = 3$ no longer cycle occurs, and (1) follows. For $q = 4$ every longer cycle is a C_5 through s. A chosen length-3 geodesic uses at most one edge of each such C_5: if it used two edges and then left the cycle, replacing that two-edge arc by the other three-edge arc would give an induced rooted path of order 5. So each C_5 contributes two vertices to R, $R \ge 2c_L$, and $[2(D+2) - m + 3R] - S = R - 2c_L + c_3 - m \ge 0$. For $q = 2$ the root is universal and $S = 2D - m$.

Induction on n. Let E be the set of endpoints of rooted induced paths of order q and z a farthest vertex, so $z \in E$. If some $x \in E \setminus \{z\}$ exists, delete it: $x \notin N[s]$, x is not a cut vertex, its loss is at most c_q, the instance remains terminal with the same q, D and m, and R drops by one. If $E = \{z\}$, delete z: the new rooted maximum is $q-1$, the eccentricity $q-2$, R is unchanged and the loss is at most 2; induction gives $S \le 2(D+q-3) - m + c_{q-1}R + 2 \le 2(D+q-2) - m + c_qR$. Deletion preserves membership in $\mathcal{B}$. This proves (1); the bipartite-or-triangle variant is identical.

(2) Attach a new path with $q-1$ edges at s. The result is terminal of index q, lies in $\mathcal{B}$, has order $n+q-1$, mass $S + 2(q-1)$ and root degree $D+1$, and m is unchanged. Substituting into (1) gives (2); since $c_q \le q-1$, ROOT(q) follows.

(3) By Theorem A.2, $G = C_{2r}$ or G contains an admissible flower $F_{L,t}$, $L = 2s$, $s + t = r$, with (S1)–(S5); we use the decomposition and notation of the proof of Theorem 8.2. In $\mathcal{B}$ the block containing the even cycle C is C itself or a bipartite block; a vertex with a consecutive-pair trace would put a triangle into that block, so there are no consecutive-pair traces and all parents have singleton traces, and the auxiliary rooted graphs used to bound the components hanging from the flower stay in $\mathcal{B}$. Part (2) with leading coefficient t bounds one parent parity class, the other costs at most $2|J|$, and (1) applied to the selected tip components J_i gives a saving of $(t-1)(t-2)$ each. Summing, $S \le (t+2)n - Lt(t+1)$, and with $n \ge L(t+1)$ the deficit from $(r-1)n$ is at least $(s-3)n + 2st(t+1) \ge 2s(r-s+1)(r-3)$. For $t \ge 2$, that is $3 \le s \le r-2$, the function $2s(r-s+1)$ is concave in s with endpoint values $6(r-2) > 4(r-1)$, so the deficit exceeds $4(r-1)(r-3)$. For $t = 1$, $R = \varnothing$ by (S4), $S \le 2n$ and $n \ge 2L = 4(r-1)$, so the deficit is at least $(r-3)n \ge 4(r-1)(r-3)$. Equality forces $n = 4(r-1)$, so $V(G) = V(F_{L,1})$; the arm vertices have distinct singleton traces, so by (S2) there are no further edges and $G = F_{2r-2,1}$, the cycle C_{2r-2} with one pendant vertex at each cycle vertex, which indeed has $S = 2n$. Finally $4(r-1)(r-3) > 2r(r-3)$ for $r \ge 5$, and C_{2r} attains $2r(r-3)$. □

Lemma 9.2 (endpoint blockers; proved)**.** *Let G be strong C_4-free and triangle-free, and let $x_0x_1\cdots x_k$ be an induced path that cannot be extended at x_k (for instance a longest induced path). If $d(x_k) \ge 2$, then $k \ge d(x_k) + 1$. If moreover G is bipartite, then $k \ge 2d(x_k)$.*

Proof. Let $y \ne x_{k-1}$ be a neighbour of x_k. It is not on the path (the path is induced) and, since the path cannot be extended, y has a neighbour x_i with $i \le k-1$. Here $i \ne k-1$ (a triangle) and $i \ne k-2$ (the 4-cycle $x_{k-2}x_{k-1}x_ky$), so $i \le k-3$. A vertex x_i is adjacent to at most one neighbour of x_k (Lemma 2.1(1)). Hence the $d(x_k) - 1 \ge 1$ such y need distinct blockers among $x_0, \ldots, x_{k-3}$, and $k - 2 \ge d(x_k) - 1$. If G is bipartite, y has the colour opposite to x_k, so its blocker x_i has the colour of x_k, that is $i \equiv k \pmod 2$ and $i \le k-4$; there are $\lfloor k/2\rfloor - 1$ such positions, so $\lfloor k/2\rfloor \ge d(x_k)$. □

Proposition 9.3 (proved)**.** *Every connected regular bipartite strong C_4-free graph satisfies* GEN_K *for every K and REAL-133. Every Moore graph (a d-regular graph of diameter 2 and girth 5) satisfies REAL-133.*

Proof. Let G be d-regular. If $d = 1$ then $G = K_2$, with $r = 1$, $\ell = 1$, $p = 2$ and $S = n$, and both statements hold. Let $d \geq 2$. As G is triangle-free, $\lambda \equiv d$ and $\ell = d$. Lemma 9.2 applied to a longest induced path gives $p \geq 2d + 1$, and Theorem 2.4 gives $p \geq 2r - 1$; so $p \geq r + d$ both when $r \leq d + 1$ and when $r \geq d + 1$. If $p \leq 2K$, then $2d + 1 \leq 2K$, so $S = dn < Kn$; thus GEN_K holds. A Moore graph of degree $d \geq 2$ is triangle-free and strong C_4-free with $r = 2$ and $\ell = d$, and the first part of Lemma 9.2 gives $p \geq d + 2 = r + \ell$. □

Remark 9.4. A Moore graph of degree 57, if it exists, has $n = 57^2 + 1 = 3250$ and $S = 57n$. It satisfies REAL-133 by Proposition 9.3, which only needs $p \geq 59$. The statement GEN_{57} imposes nothing on it (its conclusion $S \leq 57n$ holds with equality), but GEN_{56} does: since $S > 56n$, GEN_{56} requires $p \geq 2 \cdot 56 + 1 = 113$. We do not know whether such a graph would have $p \leq 112$. If it did, it would refute GEN_{56} while satisfying REAL-133; so it *could* separate GEN_K from REAL-133, but its existence alone would refute neither.

9.2. **Bipartite graphs and** GEN_5**.**

Quarantined statement 9.5 (bipartite GEN_5; payload absent)**.** (1) *Let G be bipartite, strong C_4-free and P_{11}-free, with colour classes X and Y. The recorded sentence is that G is empty whenever every vertex of X has degree at least 4 and every vertex of Y has degree at least 3.*

(2) *The recorded inequality, for every bipartite strong C_4-free P_{11}-free graph with $n \geq 2$ and colour classes of sizes a and b, is $S = 2e \leq 5n - 8 - |a - b|$. That sentence would be bipartite* GEN_5*. It is not claimed.*

Proof outline. (1) The degree assumptions force an induced P_9 starting in X (a written argument with a two-record finite check). An extension certificate then starts from this P_9: each record is a labelled induced graph with a vertex below its required degree, and lists a child for *every* admissible trace of a new neighbour (all subsets of the appropriate colour class containing that vertex, subject only to strong C_4-freeness and P_{11}-freeness). The certificate has 1030 records (436 terminal), 1029 arcs and largest order 27; every arc raises the order by one. (2) Property (1) is hereditary. Repeatedly delete an X-vertex of degree at most 3 or a Y-vertex of degree at most 2 (taking X to be the smaller class), charging 3 or 2 edges; the last two vertices span at most one edge, which leaves enough unused allowance for the bound. □

The same deletion charging would extend the bipartite bound to clique-sums: if G is strong C_4-free and P_{11}-free and is assembled from connected bipartite pieces of order at least 2 and t triangles by s_1 identifications along a vertex and some identifications along an edge, with imbalances d_i of the bipartite pieces, then $S(G) \leq 5n - 8 - 4t - 3s_1 - \sum_i d_i$ would follow from Quarantined statement 9.5. That statement is not a theorem, so neither is the clique-sum bound. The P_{12}-free analogue of the bipartite inequality fails (the incidence graph of the affine plane of order 3).

10. The band $5 < \ell \leq 6$

Quarantined statement 10.1 (two cells; not a theorem)**.** *If Quarantined statements 7.6, 6.1, 7.2 and 7.5 were theorems, then a counterexample to REAL-133 with $5 < \ell \leq 6$ would have $(r, p) = (4, 9)$ or $(r, p) = (5, 10)$. Those inputs are not independently checkable from this archive, so the dichotomy is not claimed. REAL-133 remains open.*

Proof. Here $K = 5$. Alternative (b) of Theorem 4.1 ($r = 6$, $p = 11$) is excluded by Theorem 7.6. In (a), a violation needs $p \leq r + 5$ and $p \geq 2r - 1$. For $r \leq 3$, $p \leq 8$ and GEN_4 gives $\ell \leq 4$. For

$r = 4$, $p \in \{7, 8, 9\}$; $p = 7$ is Theorem 7.2 and $p = 8$ is GEN_4. For $r = 5$, $p \in \{9, 10\}$, and $p = 9$ is Theorem 7.5. □

Both cells would follow from GEN_5. They are not closed for bipartite graphs in this paper, because Quarantined statement 9.5 is not a theorem. Neither cell is closed in general. Appendix C records partial closures whose certificates are not in this repository. Those closures are not theorems, and they are not part of the claim ladder.

11. Negative results

All examples below are explicit graphs. They were checked by direct computation, and the arguments are given.

Example 11.1 (the identity $e = n + T$ fails for $r \geq 6$). Let $s \geq 3$, $t \geq 3$ and $m \geq 1$ be integers. Let $G_{s,t,m}$ be $C_{2s} = c_0 \cdots c_{2s-1}$ with a pendant path of t edges at every cycle vertex, together with m internally disjoint paths c_0xzyc_1 (*ears*). Then $n = 2s(t+1) + 3m$, $e = 2s(t+1) + 4m$ and $T = 0$, so $e - n - T = m \geq 1$; for $(s, t, m) = (3, 3, 1)$, $n = 27$, $e = 28$, $r = 6$, $p = 11$. These graphs satisfy REAL-133 ($\ell = 2e/n < 8/3$), but they show that Theorem 7.5 has no direct analogue for $r \geq 6$, because $r(G_{s,t,m}) = s + t$ and $p(G_{s,t,m}) = 2(s + t) - 1$.

Proof. Two vertices have at most one common neighbour, so G is strong C_4-free, and it is triangle-free. An ear has length $4 > d(c_0, c_1)$, so it shortens no distance between old vertices; hence the eccentricities of the flower vertices are unchanged, and c_0, which is within distance 2 of every ear vertex, still has eccentricity $s + t$. An ear vertex w reaches the tip z of the arm at c_s only through c_0 or c_1, so $d(w, z) \geq 1 + \min\{d(c_0, z), d(c_1, z)\} = s + t$. So $r = s + t$. For the path order, let B be the block formed by C and the ears. An induced path P meets B in a subpath Q, and can leave B only at an end of Q that is a cycle vertex, through at most t arm vertices there. Split according to $Q \cap \{c_0, c_1\}$. If Q avoids both, Q lies in $c_2 \cdots c_{2s-1}$ or inside one ear, so $|Q| \leq 2s - 2$. If Q contains exactly one of them, say c_0, then $Q - c_0$ consists of at most two pieces, each inside one ear (≤ 3 vertices, ending at an ear vertex) or inside $c_{2s-1} \cdots c_2$; so either $Q \subseteq C$ with $|Q| \leq 2s - 1$, or $|Q| \leq 3 + 1 + (2s - 2)$ with at most one cycle end, or $|Q| \leq 7$ with no cycle end. If Q contains both, they are consecutive on Q, each side continues into an ear (≤ 2 vertices) or into the cycle, and the same three possibilities occur with $|Q| \leq 2s - 1$, $|Q| \leq 2s + 2$ and $|Q| \leq 6$. Hence $|P| \leq \max\{2s - 1 + 2t,\ 2s + 2 + t,\ 7\} = 2s + 2t - 1$, using $t \geq 3$ and $s \geq 3$; C minus one vertex together with the two arms at its ends attains this.

Example 11.2 (star bounds fail for $K = 5$). The natural analogue of step (iii) in the proof of Theorem 6.1 for $K = 5$ ("$L \geq 6$ and P_{11}-free exclude an induced $K_{1,6}$") is false. Two Petersen graphs sharing a vertex give $n = 19$, $S = 60$, a triangle-free strong C_4-free graph with $\min L = 6$, $p = 9$ and $\lambda = 6$ at the shared vertex. For $b \geq 2$, gluing b Petersen graphs along a common edge gives a 2-connected graph B_b with $n = 8b + 2$, $S = 28b + 2$, $\min L = 6$, $p = 9$ and $\lambda = 2b + 1$ at the two hubs (B_1 is the Petersen graph, with $p = 5$). So singleton losses cannot bound λ. These graphs violate the set-loss condition of a minimum counterexample (the eight non-hub vertices of one Petersen block have loss $28 < 5 \cdot 8 + 1$), so they are not counterexamples to GEN_5.

Example 11.3 (endpoint inequalities). Let G be connected and strong C_4-free and let (u, v) be an oriented edge. Consider the induced paths whose first two vertices are u, v in this order; let q be their maximum order and E the set of last vertices of those of order q. The *endpoint inequality* $\mathrm{EP}(q)$ asserts $D_G(E) \leq q|E|$ for all such $(G, (u, v))$ with maximum q. For a root s with $p_s(G) = q$ let E be the set of last vertices of the induced paths of order q starting at s; the rooted variant $\mathrm{EP}^+(q)$ asserts $D_G(E) \leq (q - 1)|E|$. The equivalence $\mathrm{EP}(q) \Leftrightarrow \mathrm{EP}^+(q - 1)$ for $q \geq 4$ is proved in the sense of a short formal manipulation of the definitions; the manipulation is the observation that an oriented first edge at a root is a rooted path, and conversely. The sentence that $\mathrm{EP}^+(6)$

holds is a 265-record discharging argument whose certificate is not in this repository. It is not a theorem, and it is not used to claim EP(7).

EP(9) fails. Let G have vertices $0,\dots,14$ and adjacency lists

0: 4, 8, 11	1: 5, 8, 10	2: 6, 8, 9	3: 7, 10
4: 0, 9, 10	5: 1, 9, 11	6: 2, 10, 11	7: 3, 11
8: 0, 1, 2, 12	9: 2, 4, 5	10: 1, 3, 4, 6, 13	11: 0, 5, 6, 7, 14
12: 8, 13, 14	13: 10, 12	14: 11, 12.	

It is triangle-free and strong C_4-free, with $n = 15$, $e = 23$, $S = 46$, graph6 code `N?‘@?b_JAsOw?G?A_?o`. For the oriented edge $(12, 8)$ the maximum order is $q = 9$, attained only by $12, 8, 2, 9, 4, 10, 3, 7, 11$ and $12, 8, 2, 9, 5, 11, 7, 3, 10$; so $E = \{10, 11\}$ and $D_G(E) = 20 > 18$. $\mathrm{EP}^+(8)$ fails on the subgraph induced by $0,\dots,11$ (graph6 `K?‘@?b_JAsOw`) with root 8: $p_8 = 8$, $E = \{10, 11\}$ and $D(E) = 16 > 14$. Componentwise charging versions fail as well. EP(8) is open.

Example 11.4 (the Petersen graph refutes a dominating-core budget). Let $k = p - r$. Call an induced subgraph H of G an *eligible core* if it is proper, connected, dominating, and $p(H) \le p(G) - 2$ (Theorem 2.5 supplies such cores, apart from a cycle alternative). The budget CB asks for an eligible core with $S(G) - S(H) \le k(n - |H|) + |H|$; together with REAL-133 for H by induction it would imply REAL-133. The Petersen graph has $S = 30$, $p = 5$, $r = 2$, $k = 3$. An eligible core has $p(H) \le 3$; a connected triangle-free strong C_4-free P_4-free graph is a star; a star in the Petersen graph with $k \le 3$ leaves dominates at most $4 + 2k$ vertices, which is 10 only for $k = 3$, so the eligible cores are exactly the ten closed neighbourhoods $N[v]$. Each has $S(H) = 6$ and $S(G) - S(H) = 24 > 3 \cdot 6 + 4 = 22$. Among all 28,003,330 eligible cores of connected strong C_4-free graphs with $n \le 12$, only these ten fail. A repaired version restricted to $\ell > 5$ remains open (Conjecture 13.3).

Example 11.5 (a scalar relaxation that does not imply the conjecture). The degree, triangle and distance data of the Heawood graph ($n = 14$, cubic, $T = 0$, $r = 3$, $S = 42$, $Z = 28$) satisfy all the identities used in our counting arguments, including the Moore-type identity $\sum_v d(v)^2 = n(n-1) + 6T - 2Z$ (Z the number of nonadjacent pairs without a common neighbour), together with the generic lower bounds $p \ge 2r-1$ and $p \ge \min_v \lambda(v)+2$, if p is replaced by the value 5. But $5 < r + \ell = 6$. So this particular system of scalar constraints together with these two path lower bounds does not imply REAL-133. This does not exclude arguments that use further inequalities relating degrees, eccentricities and triangles to p. (The real Heawood graph has $p = 7$.)

Example 11.6 (radius-extremal graphs with large ℓ). Let F be the incidence graph of the projective plane $\mathrm{PG}(2, q)$, with $f = |F| = 2(q^2+q+1)$ and $S(F) = (q+1)f$, rooted at a vertex. Let $t = f-1$ and $b \ge 1$. Take $C_6 = c_0 \cdots c_5$, attach a pendant path with t edges at each cycle vertex, and attach b copies of F at c_0 by identifying their roots with c_0. The result is strong C_4-free (each new block meets the rest in one cut vertex) and

$$n = 6f + b(f-1), \qquad S = 12f + b(q+1)f, \qquad r = t + 3, \qquad p = 2t + 5 = 2r - 1.$$

Indeed the distances inside the flower are unchanged; every vertex of an attached copy is within distance $f - 1 \le t$ of c_0, so each cycle vertex has eccentricity $t + 3$ and the flower already has radius $t + 3$, while a vertex x of a copy has distance $d_F(x, c_0) + t + 3$ from the tip of the arm at c_3. An induced path meets the cycle in at most five consecutive vertices, and each of its two ends outside the cycle lies in one arm or one copy of F, with at most t vertices; the flower contains an induced path on $2t+5$ vertices. Mass is additive at cut vertices. For fixed q, $\ell \to (q+1)f/(f-1)$ as $b \to \infty$; letting then $q \to \infty$, with $r = f + 2 \sim 2q^2$, this limit is $\sim \sqrt{r/2}$. For example $q = 5$, $b = 17$ gives $n = 1409$, $r = 64$, $p = 127$ and $S = 7068 > 5n$. These graphs satisfy $\mathrm{RAD}(r)$, but they show that ℓ is unbounded among radius-extremal graphs, so no absolute bound such as $\ell \le 5$ holds for all r, and $\mathrm{RAD}(r)$ needs a bound depending on r.

Further refuted routes (path-count recurrences, random induced walks, ball-deletion lemmas, and two approaches to TD) are recorded in Appendix D.

12. Relation to earlier work

Three published theorems are used as black boxes and are stated in Section 2. Erdős, Saks and Sós proved that every connected graph has an induced path on at least $2r(G)-1$ vertices [4]; together with the elementary bound $n \geq 2r$, this is the reduction of Theorem 4.1. Camby and Schaudt characterised the absence of a long induced path by a connected dominating set whose induced subgraph is a shorter path-free graph or a cycle [1]; the radius-extremal arguments and the cell analysis apply that theorem and then classify traces. Seinsche's theorem on graphs with no induced P_4 [10] is used only for the small-mass case $\ell \leq 3$.

The conjecture itself is item 133 of DeLaVina's list of Graffiti.pc conjectures [3]. We follow the real-valued reading fixed in the Formal Conjectures repository [5, 6]. Chen [2] studies mainly the earlier floor form $p \geq r + \lfloor \ell \rfloor$. The comparison with our Theorem A is in the introduction. We have not used the arguments of that preprint, and our proofs do not depend on it.

13. Conjectures and open problems

Theorem 4.1 reduces a counterexample of REAL-133 to a violation of some GEN_K or some $\mathrm{RAD}(r)$. Theorem 3.7 settles $\ell \leq 3$ and the case $r \leq 2$, $\ell \leq 4$. The inputs that would settle $3 < \ell \leq 5$ are quarantined and are not theorems. Unrestricted GEN_6 and REAL-133 are open. The following are open as well.

(1) **REAL-133 for $3 < \ell \leq 6$.** Quarantined statement 10.1 would place a counterexample with $5 < \ell \leq 6$ in one of two cells if its hypotheses were theorems. They are not. Appendix C does not close those cells.

(2) **GEN_K for $K \geq 3$ on the claim ladder, and GEN_K for $K \geq 5$ in any form.** GEN_3 and GEN_4 are quarantined. On one endpoint route, the missing steps include EP(8) and a boundary-selection inequality for $K = 5, 6$; the report that states that inequality is not in this repository. EP(9) fails (Example 11.3), so that route cannot be uniform in q.

(3) **$\mathrm{ROOT}(q)$ for $q \geq 4$ as a theorem, and $\mathrm{RAD}(r)$ for $r \geq 4$ as a theorem.** $\mathrm{ROOT}(3)$ is proved. The cases $4 \leq q \leq 8$ and $4 \leq r \leq 10$ are quarantined. For $q = 9$, all 350 seeds of $\mathcal{L}_9$ remain open, and $\mathrm{RAD}(r)$ is open for every $r \geq 11$.

(4) **Level 6 and REAL-133.** Partial Result G does not empty the class of triangle-free strong C_4-free graphs with $p \in \{11, 12\}$, minimum degree at least 4, and every endpoint of every longest induced path of degree at most 7. Whether every graph in that class has a nonempty 6-cheap set is open. So is $S \leq 6n$ for every triangle-free strong C_4-free graph with $p \leq 12$ (unrestricted triangle-free GEN_6), and so is the same bound when triangles are allowed. The inductive passage from GEN_{K-1} to GEN_K is open for the native path cap $2K$ when $K \geq 7$. REAL-133 is open. Appendix E proves two port lemmas and an averaging inequality, gives one counterexample, and quarantines the remaining notes. Lemma E.5 shows that one proposed quotient on the graph G_{41} is not proper. It does not assert that no positive proper model of G_{41} exists, and it does not prove unrestricted GEN_6.

Conjecture 13.1 (uniform rooted and radius bounds)**.** $\mathrm{ROOT}(q)$ *holds for every $q \geq 3$, and for every $r \geq 5$ the bound $S \leq (r-1)n - 2r(r-3)$ of Theorem 8.2 holds, with equality only for C_{2r}.*

The case $q = 3$ is Lemma 5.2. The cases $4 \leq q \leq 8$ and $5 \leq r \leq 10$ are quarantined, not proved. In the class $\mathcal{B}$, $\mathrm{ROOT}(q)$ holds for every q by Theorem 9.1(2); the radius form in $\mathcal{B}$ is part (3) of that theorem and is not claimed. The conjecture would follow from the next statement if that statement were proved.

Conjecture 13.2 (terminal correction TD)**.** *If F is connected and strong C_4-free, $p_s(F) = q$ and* $\mathrm{ecc}(s) = q - 1$*, then*

$$S(F) \le (q-1)(n-1) - (q-3)D - m - (q-2)(q-3).$$

Equivalently, $2\beta - (T - m) \le (q-3)\,|V \setminus (P \cup N[s])|$ for a root-to-farthest geodesic P, where $\beta = e - T - n + 1$. A derivation that TD implies ROOT(q) for all q, and hence Conjecture 13.1, is not written in this paper, so that implication is not a theorem here. The terminal bound in $\mathcal{B}$ is Theorem 9.1(1). A count of 3,646,151 terminal roots on connected strong C_4-free graphs with $n \le 14$, not re-runnable from this archive, recorded no violation of TD. That count is not a proof.

Conjecture 13.3 ($\mathrm{CB}_{\mathrm{high}}$)**.** *Every connected strong C_4-free graph G that is not a cycle and has $\ell(G) > 5$ has an eligible core H with $S(G) - S(H) \le (p-r)(n - |H|) + |H|$.*

If $\mathrm{CB}_{\mathrm{high}}$, the range $\ell \le 5$, and an order induction were all available, REAL-133 would follow with no further GEN_K. Quarantined statement 8.3 does not supply the range $\ell \le 5$, so the conjunction does not prove REAL-133. There is no test data in this archive: no connected strong C_4-free graph with $n \le 12$ and $\ell > 5$ is exhibited. Example 11.4 shows that the restriction $\ell > 5$ is necessary for the budget as stated, and Example 11.5 shows that a proof cannot rely on the scalar constraints listed there alone.

14. Conclusion

The claim ladder of this paper is Theorem 3.7, Theorem 4.1, Corollary 7.3, the rooted bound ROOT(3), the rooted bounds of every index in the class of graphs whose blocks are bipartite or cycles, the regular bipartite and Moore cases, the explicit counterexamples of Section 11, and the three reductions of Section 1. In particular REAL-133 holds for $\ell \le 3$, for $r \le 2$ with $\ell \le 4$, and for $r \ge 5$ with $\ell \le 4$.

GEN_3, GEN_4, ROOT(q) for $4 \le q \le 8$, RAD(r) for $4 \le r \le 10$, bipartite GEN_5, Struct(r), and the verdict records (M), (S) and (P) are not independently checkable from this archive. They are not theorems. The deduction that would give REAL-133 for $\ell \le 5$ is written and is not claimed. Partial Result G leaves the residual class $\mathcal{R}_G$ open. Proposition 1.4 does not show that the residual is empty, and it does not prove unrestricted triangle-free GEN_6. Unrestricted GEN_6, RAD(r) for every $r \ge 11$, and REAL-133 remain open.

15. Method: computer assistance and verification

The results of Sections 5–11 and of the appendices were obtained in rounds. In each round the AI model GPT 6 Astra worked on a precisely stated target in a fresh session, with a context file listing the definitions and the previously accepted results, and wrote a report together with its certificates as JSON files. A claim was accepted only after a second, separate GPT 6 Astra session, acting as referee, had checked it. In this paper "referee" and "independent" refer to such separate sessions of the same model; they are not external peer review and not an independent human assessment. The referee session shared no session state with the prover beyond the claim, the prover's report and the certificate files. It was instructed to be adversarial, to recompute the finite certificates with code it wrote itself (without running the prover's code as a substitute), and to return a verdict CORRECT, CORRECT WITH FIXES, INCORRECT or UNVERIFIED for each claim. A claim judged INCORRECT is not used as a hypothesis. A claim judged CORRECT or CORRECT WITH FIXES is, in this paper, only a verdict record. It is not a hypothesis of any proved theorem, because the programs and search payloads are not in this repository. Separate sessions of one model can share systematic failure modes. The stand-alone checkers and the Lean development that were used in earlier rounds are not files in this archive, so they do not make GEN_3, GEN_4, or the later enumerations independently checkable here.

The referees did reject or correct claims. Examples: a lemma in the cactus case of TD was false as stated (a 7-vertex counterexample) and was repaired; a reported count of 4,881 maximal parent families was in fact 2,218 (search leaves had been counted); a rooted-P_5 statement proved only for $n \leq 12$ was demoted to a conjecture until a later round proved it; several statements needed explicit hypotheses (ambient strong C_4-freeness, actual parents meeting the component, $t \geq 2$) without which they are false, and these hypotheses are included above. Search totals from runs that could not be replayed, and statements about the implementation history that the referees could not authenticate (marked UNVERIFIED in the verdict files), are not used in any proof and are not presented here as independently checked; where an original script was missing, the referee's replacement computation is the one that was accepted in that round. Those programs are not files in this repository (Section 16).

The certificate types described in the quarantined outlines are extension DAGs, trace classifications, and exclusion lists. Those files are not in this repository. Completeness of an enumeration cannot be checked from the JSON verdict alone. No computer-assisted inequality in this paper is formally verified from the archive. The Lean development of GEN_3 is absent.

Partial Result G separates three proved reductions (Section 1) from enumerated verdict records in rounds R35–R40. The search payloads behind those records are absent, so the records are not theorems and $\mathcal{R}_G$ stays open. In Appendix E the port lemmas and the averaging inequality are proved, Lemma E.5 is a counterexample to one quotient, and every other note is quarantined. Lemma E.5 shows that a prescribed one-vertex quotient is not proper. It is not a proof that no positive proper model exists, and it is not used as a hypothesis anywhere else. Unrestricted GEN_6 and REAL-133 are not theorems.

16. Data availability and reproducibility

This section states which verification files the public repository contains, how to check them, and which claims remain conditional on files that are not included.

Files present. The directory `certificates/` contains one JSON file `ROUND_BRANCH_R`k`_CERTIFICATE.json` for each round $k \in \{1, \ldots, 40\} \cup \{46, \ldots, 53\}$, together with a `sha256sum` sidecar of the same stem ending in `.sha256`, and a second copy of that sidecar ending in `.json.sha256`. There are 48 JSON files. Rounds R41–R45 are not included. There is no R54 file. The notes `notes/RESULTS_TO_2026-10-24.md` and `notes/RESULTS_TO_2026-11-01.md` summarise referee headlines. They are not proofs. The directory `data/` contains only `data/README.md`, which points back to `certificates/`.

Appendix F lists every JSON file with its SHA-256 value. The records used by Partial Result G and by Appendix E are the following.

File	SHA-256
R35	132f7ecaba1663f20811bb8930e71f55ac481aa55003d638ca2ad3c0841eb0af
R36	410e10c00646bc3cba711a601b826b83a2078c4fec0b474ab0340e2c91137b34
R37	27894d51a027f033ec005b94f6ff57f75c1a4bdb866eb926af207324b6683758
R38	0f199e4ec5681d92467b04bd0747029235edba3341ef7b3f3f13feeec2ee2bd5
R39	dc1224bcc8843e6fdbcf285e68d036e0945165acb585b1c8f65c5bec5bb34477
R40	71819fcdd5681a6cb5413a6ff326d1d6d651ebd5f2e316886d8f4abc0de9e4f7
R51	aebe0c00de6fd8ed0059280a2ac6d9c56e6c64355487912d0fde5a5be04999f7
R52	8383d17dac2d9f23995b066b4a9e82a6b977602ee9451bab90fd810161ee171a
R53	b90cc797e755cb8cf849937c20ad1576ea551a7b0ec3b780676c3e84d81b6ecd

Each abbreviated name Rk in that table denotes the file `certificates/ROUND_BRANCH_R`k`_CERTIFICATE.json`.

How to check. From the repository root, the following commands compare each JSON file with its sidecar. The sidecar format is one line, `HASH` followed by two spaces and the file name, as produced by `sha256sum`.

```
cd certificates
sha256sum -c ROUND_BRANCH_R40_CERTIFICATE.sha256
sha256sum -c ROUND_BRANCH_R53_CERTIFICATE.sha256
for f in ROUND_BRANCH_R*_CERTIFICATE.sha256; do
  sha256sum -c "$f" || exit 1
done
```

A successful check shows that the JSON bytes are the bytes named in Appendix F. It does not re-run a graph search, does not execute a checker program, and does not recreate a referee session. What the JSON establishes is the stored verdict of each listed claim, including the claim identifier, the recorded statement, and the mark CORRECT, CORRECT WITH FIXES, INCORRECT or UNVERIFIED. Matching the hash authenticates that record. It does not authenticate a file the JSON only mentions.

The R46 file records, as an internal pointer, an R45 hash `1b05993f54e0fb96942dd0ef9af0c2529a6e731ea11cc7d243a6d8a4a8136b33`. That hash is not the name of a file in this repository, and no argument below treats it as a checked input except where a lemma is explicitly qualified by the missing file.

Files not present, and the claims they qualify. The following paths are cited by the article or by fields inside the JSON files, and they are not in this repository:

- `data/rounds/`, including `check_lemmas.py`, the programs `verify_round_branch_r1.py` through `r6.py`, the lane reports under `lanes/`, the referee reports under `verdicts/`, the Lean development, and the extension-DAG payloads;
- certificate JSON for rounds R41–R45 and for R54;
- the prover directories named inside the later certificates.

The consequences for the claim ladder are as follows.

- Quarantined lemmas 5.6–5.8, Quarantined statements 6.1, 8.1, 7.5–8.2, 9.5 and A.2, and the finite parts of Appendix C are not re-runnable from this repository. They are not theorems. Rounds R1–R10 contribute JSON records of accepted counts. They do not contain the programs that produced the counts. Rounds R41–R45 and R54 are absent. In particular every argument that cites R45 is not checkable here.
- The proved reductions of Section 1 do not use the R35–R40 searches. The lists (M), (S) and (P) are verdict records only. Their search payloads are absent, so those lists are not on the claim ladder, and the residual class $\mathcal{R}_G$ is open.
- Quarantined lemma E.6 depends on the missing round R45. Quarantined lemmas E.3, E.7 and E.8 depend on enumerations or constructions that are not written in this paper. None of them is a theorem.
- No absent file is used to assert unrestricted GEN_6 or REAL-133. Both remain open. A successful `sha256sum` check authenticates JSON bytes. It does not promote a quarantined statement to a theorem.

Acknowledgments

The authors used AI models and AI tools in this work, as follows. The AI model GPT 6 Astra produced, in separate prover sessions, the proofs of the local lemmas of Section 5, of GEN_4 and of the results of Sections 7–11 and the appendices, together with their finite certificates and the counterexamples; in earlier work it also suggested the hand proof of Fact 2 in Lemma 3.6, the equality case of Lemma 3.6, and the argument of Theorem 7.2 with the classification in Lemma 7.1.

Separate GPT 6 Astra sessions acted as referees: they checked every accepted result, recomputed the certificates with their own code, and supplied several repairs; GPT 6 Astra sessions also reviewed drafts of this manuscript. The AI model GPT 6.1 Sol, run through OpenAI Codex, was used as a simulated referee of earlier versions of the manuscript. Grok Bot, an AI agent tool, was used to orchestrate the sessions, to run computations and the stand-alone checkers, to produce the Lean formalisation of GEN_3, and to draft and edit the text. OpenAI Codex was used as the tool through which the GPT 6 Astra and GPT 6.1 Sol sessions were run. The authors are responsible for the text, including any errors. Neil Xu's correspondence address is omitted from this version.

Appendix A. The structure theorem Struct(r)

Let $L = 2s \geq 6$ and $t \geq 1$. The *flower* $F_{L,t}$ consists of a cycle $C = c_0c_1 \cdots c_{L-1}c_0$ and, for every i, an arm $c_i = v_{i,0}, v_{i,1}, \ldots, v_{i,t} = z_i$; indices of cycle vertices are taken modulo L, and $\delta(i,j)$ denotes the distance of c_i, c_j on C. For a vertex $x \notin V(F)$ its *trace* is $N(x) \cap V(F)$. Write $S_{i,j} = \{v_{i,j}\}$, $E_{i,j} = \{v_{i,j}, v_{i,j+1}\}$ and $B_i = \{c_i, c_{i+1}\}$.

Definition A.1. An induced copy of $F_{L,t}$ in a graph G is *admissible* if G is strong C_4-free and P_{L+2t}-free and

$$d_G(c_{i+s}, z_i) \geq s + t \qquad (0 \leq i < L). \tag{3}$$

For an admissible flower put $A = N(C) \setminus V(C)$ and $R = V(G) \setminus (V(C) \cup A)$; the *parents* of a component J of $G[R]$ are its neighbours in A, and $J + a = G[V(J) \cup \{a\}]$. For $r \geq 5$, Struct(r) is the statement: *every connected strong C_4-free graph G with $r(G) = r$ and $p(G) = 2r - 1$ is C_{2r} or contains an admissible $F_{2s,t}$ with $s + t = r$, $s \geq 3$, $t \geq 1$, such that*

- (S1) every vertex outside the flower has trace $\varnothing$, $S_{i,j}$ or $E_{i,j}$ $(0 \leq j < t)$, or B_i; in particular every vertex of A has a singleton or consecutive-pair trace on C;
- (S2) every edge of $G[A]$ joins two vertices with the same singleton trace on C, and these edges form a matching;
- (S3) for every component J of $G[R]$ and every parent a of J, every induced path of $J + a$ starting at a has at most t vertices;
- (S4) if $t = 1$, then $R = \varnothing$;
- (S5) every vertex of R has at most two neighbours in A; if it has two, it has empty trace, and its two parents are nonadjacent with traces $S_{i,0}, S_{i+1,0}$ for some i (this occurs only when $t \geq 3$). Consequently $A = A_0 \cup A_1$, where A_ϵ contains the vertices with singleton trace c_i, $i \equiv \epsilon \pmod 2$, and the vertices with a pair trace are put into either class, and every vertex of R has at most one neighbour in each class.

The classification of adjacent pairs of outside vertices is also part of the record: the admissible unordered trace pairs of two adjacent outside vertices are exactly $(\varnothing, \varnothing)$, $(S_{i,j}, S_{i,j})$ for $j < t$, $(\varnothing, S_{i,j})$ and $(\varnothing, E_{i,j})$ for $j \leq t - 2$, $(\varnothing, B_i)$ for $t \geq 2$, $(S_{i,j}, S_{i,j+3})$ and $(E_{i,j}, S_{i,j+3})$ for $j \leq t - 4$, and $(S_{i,j}, E_{i,j+2})$ for $j \leq t-3$; there are $1+L(2t+1)$ single traces (including $\varnothing$) and $1+L$, $1+5L$, $1 + L(6t - 9)$ adjacent pairs for $t = 1$, $t = 2$, $t \geq 3$.

Quarantined statement A.2 (Struct(r); payload absent)**.**

1. *If G contains an admissible $F_{L,t}$, then* (S1)–(S5) *and the pair classification hold.*
2. Struct(5) *holds.*
3. *The step from* Struct(r) *to* Struct($r + 1$) *is only outlined below. The write-up is absent, and the base uses Quarantined statement 7.5. So* Struct(r) *is not a theorem of this paper.*

Sources and outline. (1) and (3) are proved in `lanes/astra_r5/radr/report.md`, §§1–7, and were audited in `audits_r2_r6/astra_r6_auditStruct.report.md`. The single-trace classification rests on the observation that, for a vertex adjacent to $v_{i,a}$ and $v_{j,b}$, (3) is preserved exactly

when $|a-b| \le 2$ $(i=j)$ or $\delta(i,j)+|a-b| \le 2$ $(i \ne j)$, and that C_4-safety is also a pairwise condition; every excluded trace yields an explicit induced P_{L+2t} obtained by joining two arms through a long arc of C. An arbitrary-length bridge lemma (an induced path outside F with empty internal traces cannot join two different arms when one end has positive depth) gives (S3): an induced path $a, x_1, \dots, x_t$ with $a \in A$ and $x_j \in R$ could be completed by a cycle segment of $L-1$ vertices and a full arm to an induced P_{L+2t}. (S4) is the case $t=1$ of (S3), and (S5) follows from the pair classification and the bridge lemma. All lists were in addition recomputed for the 44 flowers with $5 \le s+t \le 12$.

(2) The proof of Theorem 7.5 gives, for a connected strong C_4-free graph of radius 5 without P_{10}, that $G = C_{10}$, or G contains C_8 with a pendant vertex w_i at each c_i, or C_6 with a pendant P_2 at each c_i, where the w_i (respectively the ends of the pendant paths) were chosen at distance 5 from c_{i+s}. These are admissible copies of $F_{8,1}$ and $F_{6,2}$, and (1) supplies (S1)–(S5).

(3) Let G have radius $r+1$ and no P_{2r+2}. Theorem 2.5 gives a connected dominating induced H that is C_{2r+2} (then $G = C_{2r+2}$ by the dominating-cycle shortcut argument) or P_{2r}-free; then $r(H) = r$ and $p(H) = 2r-1$. If $H = C_{2r}$, witnesses at distance $r+1$ from the cycle vertices have singleton traces and give arms of depth 1. Otherwise Struct(r) gives a flower in H, and a witness w_i with $d_G(c_{i+s}, w_i) \ge r+1$ lies outside H; for $t \ge 2$ each of its H-neighbours lies in a component $J \subseteq R(H)$ all of whose parents have the singleton trace c_i (for $t=1$ use (S1)–(S2) instead), by the same distance computation as below, and a shortest parent-to-neighbour path extended by w_i is an arm of depth $t+1$. Edges between new tips are excluded by (3) or by an explicit induced P_{L+2t+2}. So G contains an admissible $F_{L,t+1}$, and (1) applies. □

Selected tip components. The distance inequality (3) is what makes the flower usable for mass bounds. Let $t \ge 2$, let J_i be the component of $G[R]$ containing $v_{i,2}, \dots, v_{i,t}$, and let a be a parent of J_i with $c_j \in N(a)$. By (S3) a shortest path from a to z_i in $J_i + a$ has at most $t-1$ edges, so $d_G(c_{i+s}, z_i) \le \delta(i+s, j) + 1 + (t-1) \le s+t$, with equality only if $\delta(i+s, j) = s$, that is $j = i$. By (3), every parent of J_i has trace exactly $\{c_i\}$. In particular the L components J_i are distinct and have unique parents. These are the facts used in the proofs of Theorems 8.2 and 9.1(3). Radius is not hereditary, and cannot be imposed on the finite templates used in the classifications; inequality (3) is the hereditary substitute that the finite pruning uses.

Appendix B. The finite local criterion $\mathcal{L}_q(b)$

We use the parent systems (J, B), the boundary mass $M_B(J) = S(J) + 2e(B, J) - 3T_{B,J,J}$ and the statement C_q of Section 8; C_q is equivalent to ROOT(q). Assume ROOT$(q-1)$ and let (J, B) be a counterexample to C_q with $h = |J|$ minimum. For $\varnothing \ne U \subseteq V(J)$ let $\Delta_B(U) = M_B(J) - M_B(J - U)$, where $M_B(J-U)$ is summed over the components of $J - U$ with their actual parents.

Quarantined lemma B.1 (argument not in this archive)**.** (1) *$\Delta_B(U) \ge (q-1)|U| + 1$ for every nonempty $U \subseteq V(J)$; in particular $L_J(v) = 2d_J(v) - 3t_J(v) \ge q-2$ for every v, and (J, B) has at least $q-1$ parents.*

(2) Templates. *Let $K \subseteq V(J)$, $B_0 \subseteq B$ nonempty, and let Q be $G[K \cup B_0]$ together with an auxiliary root ρ adjacent exactly to B_0. A set $A \subseteq V(Q) \setminus \{\rho\}$ is an* admissible trace *if adding a new vertex y with $N(y) = A$ gives a strong C_4-free graph with no induced path on $q+1$ vertices starting at ρ; let $\mathcal{A}(Q)$ be the set of admissible traces meeting K. For $\varnothing \ne U \subseteq K$ let $\delta_Q(U) = 2\#\{$edges of $Q - \rho$ meeting $U\} - 3\#\{$triangles of $Q - \rho$ meeting $U\}$, let $u_0(U)$ be the number of vertices of U without a neighbour in B_0, and $c_U(A) = 2|A \cap U| - 3\#\{vw \in E(Q-\rho) : v, w \in A, \{v, w\} \cap U \ne \varnothing\}$. If every $A \in \mathcal{A}(Q)$*

with $c_U(A) > 0$ has $|A| \geq 2$, then

$$\Delta_B(U) \leq \delta_Q(U) + 2u_0(U) + \sum_{A \in \mathcal{A}(Q)} \max\{0, c_U(A)\}.$$

(3) Raising rule. *If Q is embedded as an induced subgraph in (J, B) and $v \in K$ satisfies $L_K(v) - 3T_{B_0,K,K}(v) < q - 2$, then v has a neighbour $y \in V(J) \setminus K$ whose trace on Q is admissible, contains v, and misses $N_K(v) \cup \{\rho\}$.*

Definition B.2. The *seeds* $\mathcal{S}_q$ are the templates formed by $q-1$ parents, one private neighbour (child) of each, and ρ, where the children induce a strong C_4-free graph on $q-1$ vertices other than P_{q-1} (one representative per isomorphism class). A template Q *wins by deletion* if some nonempty $U \subseteq K$ satisfies the hypothesis of Lemma B.1(2) with right-hand side at most $(q-1)|U|$. For an integer $b \geq 0$ define $W_q(Q, b)$ recursively: it holds if Q wins by deletion, or if $b > 0$ and there is a vertex v as in Lemma B.1(3) such that $W_q(Q + y_A, b-1)$ holds for every admissible trace A that contains v and misses $N_K(v) \cup \{\rho\}$ (vacuously if there is none); a vertex v as in (3) with no such trace also gives a win at every depth. Otherwise $W_q(Q, b)$ is false. Finally $\mathcal{L}_q(b)$ is the statement that $W_q(Q, b)$ holds for every $Q \in \mathcal{S}_q$.

$\mathcal{L}_q(b)$ is a finite decidable statement: retained templates have at most $2q - 1 + b$ vertices.

Quarantined statement B.3 (implication whose write-up is not in this archive)**.** *For $q \geq 4$ and $b \geq 0$,* ROOT$(q-1)$ *and $\mathcal{L}_q(b)$ together would imply* ROOT(q) *if Quarantined lemma B.1 were a theorem. The write-up of that lemma is not in this repository, so the implication is not a theorem of this paper.*

Proof. A minimum counterexample to C_q has at least $q-1$ parents by Lemma B.1(1), hence contains a seed. Following the raising rule extends the induced embedding along the certificate; a deletion win contradicts Lemma B.1(1), and a raising vertex with no admissible trace contradicts Lemma B.1(3). □

The numbers of seeds are $7, 17, 43, 116, 350$ for $q = 5, \ldots, 9$. No depths b_q completing $\mathcal{L}_q(b_q)$ are known for $q \geq 9$, and the criterion is only sufficient.

Appendix C. The band $5 < \ell \leq 6$: setting and partial closures

Throughout, G is connected and strong C_4-free, in Cell A ($r = 4$, $p = 9$) or Cell B ($r = 5$, $p = 10$). To *close* a subcase means to prove $S(G) \leq 5n$ for every graph of the cell satisfying its hypotheses. All results below hold at every order. Sources are given in Appendix F.

Nested cores and subcases. Theorem 2.5 gives a connected dominating induced subgraph H (the *core*) that is C_{p+1} or P_{p-1}-free, and applying it again inside H gives an *inner core* J dominating H; then J 2-dominates G.

- **A1** (Cell A): H is P_8-free with $r(H) = 3$, and $J \subseteq H$ is a dominating P_6-free induced subgraph with $r(J) = 2$.
- **A2** (Cell A): G has an induced C_6 that 2-dominates G (for instance when $r(H) = 4$, $p(H) = 7$ and H has a dominating induced C_6 as in the proof of Theorem 7.2).
- **B1** (Cell B): $(r(H), p(H)) = (4, 8)$, $(r(J), p(J)) = (3, 6)$, and G has no induced 2-dominating C_6.

Layers for A2. For a 2-dominating induced $C = c_0 \cdots c_5$ put $A = N(C) \setminus C$, $R = V(G) \setminus (C \cup A)$ and $L = G[C \cup A]$. For a component D of $G[R]$ let $P = N(D) \cap A$ be its *actual parents*, $h = |D|$, $k = |P|$, and

$$M(D) = S(D) + 2e(P, D) - 3\tau_1 - 3\tau_2,$$

where τ_1 counts triangles with one vertex in P and two in D, and τ_2 triangles with two vertices in P and one in D. Then $S(G) = S(L) + \sum_D M(D)$. A parent is *normal* if its trace on C is a singleton or a consecutive pair, and *antipodal* if it is $\{c_i, c_{i+3}\}$; there are at most three antipodal vertices.

Layers for A1 and B1. Put $L_0 = V(J)$, $L_1 = \{v : d(v, J) = 1\}$, $L_2 = \{v : d(v, J) = 2\}$ (these are all vertices) and $H^* = G[L_0 \cup L_1]$, the *first-layer core*. A component D of $G[L_2]$ has actual parents $N(D) \cap L_1$. *Unique parents relative to* H^* means $|N(v) \cap V(H^*)| = 1$ for every $v \in L_2$. This does not follow from the corresponding condition relative to H, and $p(H) \le 8$ does not imply $p(H^*) \le 8$ (counterexamples of order $2m + 16$ and 22 in the supplement); results stated under unique parents assume the condition relative to H^* explicitly, together with any path bound on H^* that they use.

Verified partial closures.

(1) *Bipartite graphs and clique-sums.* Both cells are closed for bipartite graphs (Theorem 9.5) and for strong C_4-free P_{11}-free graphs assembled from connected bipartite pieces and triangles by vertex and edge identifications (the bound after Theorem 9.5).

(2) *Dominating cycles.* In Cell A, if G has a dominating induced C_k with $k \in \{6, 7, 8, 10\}$, then $S \le 5n$; in Cell B the same holds for a dominating induced C_9 or C_{11}, and when $p(H) = 7$ (computer-assisted).

(3) *P_6-free blocks.* Every 2-connected strong C_4-free P_6-free graph has at most 13 vertices, and there are exactly 11 of them (proved, with a finite classification).

(4) *A2, few parents* (proved). If $h \ge 2$ and $k \le 4$, then $M(D) \le 5h$; if $k \le 3$, then $M(D) \le 4h$. This follows from the identity

$$(k+1)h - M(D) = \sum_{v \in D} \big[k - w_v + (b_v - 1)(d_D(v) - 1)\big] + \tau_1 + 3T(D) + 3\tau_2,$$

where $b_v = |N(v) \cap P|$ and $w_v = |(N(v) \cap P) \cup \bigcup_{z \in N_D(v)} (N(z) \cap P)| \le k$, in which every summand is nonnegative for $h \ge 2$. The bound fails for singletons, where $M(\{v\}) = 2k - 3e(G[P])$.

(5) *A2, five normal parents* (proved, with a 63-trace finite classification on P_6). If D has at least five normal parents, then D is a connected graph whose blocks are edges and triangles, of diameter at most 4; such a D has a vertex of eccentricity at most 2 in D.

(6) *A2, global charge* (proved). Select a set $\mathcal{C}$ of residual components each of which is a singleton or an edge/triangle block tree of diameter at most 4, and assume $M(D) \le 5|D|$ for every unselected component. Let I be the bipartite incidence graph between $\mathcal{C}$ and the set Q of their actual parents. If I is a forest and every edge of $G[A]$ has a common neighbour on C, then $S(G) < 5n$. More generally $S(G) \le 5n - \Delta + 2|Q| + 2\beta(I) - 2f(I) - \eta$, where $\Delta = 5|L| - S(L)$, $\beta(I)$ and $f(I)$ are the cycle rank and the number of components of I, and $\eta \ge 0$ is an explicit sum of component savings.

(7) *A2, independent first layer* (proved). If A is independent and every residual component has at most three parents, then $S \le 5n - 12$.

(8) *Rooted-P_5 theorem* (proved). If Q is 2-connected, strong C_4-free and P_7-free and some vertex starts no induced P_5, then $r(Q) \le 2$. So every vertex of such a graph of radius 3 starts an induced P_5.

(9) *B1 inner cores with cut attachments* (proved). Let J be a 2-connected strong C_4-free P_7-free graph of radius 3, let $G \supseteq J$ be strong C_4-free and P_{11}-free, and suppose that every component D of $G - J$ is attached to J at a single vertex v_D. Then $S(G) \le 5n - 2|J| - 3\beta - \tau$, where $\beta = \sum_D |N(v_D) \cap D|$ and $\tau = \sum_D e(G[N(v_D) \cap D])$.

(10) *Minimum counterexamples to* GEN_5 *and* GEN_6 (computer-assisted). A counterexample to GEN_5 of minimum order is 3-connected, for every 3-vertex cut X the graph $G -$

X has a component consisting of one triangle-free vertex of degree 3, and every vertex starts an induced P_9. A counterexample to GEN_6 of minimum order is 4-connected, with the analogous property for 4-vertex cuts, and every vertex starts an induced P_8. A counterexample to GEN_K with radius $r \geq 2$ has $n \geq K^2 + r$.

Further closures for special attachment patterns in A1 and B1 (prefix closures with unique outer parents, hub-pair dominated blocks, maximal-block reductions) are proved in the supplement under the hypotheses stated there; we do not restate them. *Examples.* Families with $\ell \to 4$ exist in both cells, and among graphs with $n \leq 15$ the largest values of ℓ are 2.93 (Cell A) and 2.67 (Cell B). The componentwise bound $M(D) \leq 5|D|$ fails in A2: C_6 with a leaf a_i at each c_i, plus an edge xy with $x \sim a_0, a_1$ and $y \sim a_2, a_3, a_4$, has $n = 14$, $r = 4$, $p = 9$, $S = 36$, and $D = \{x, y\}$ has $M(D) = 12 > 10$; there is also an infinite three-parent family.

Open subcases. A1 with non-unique or direct attachments; A2 components with five to seven parents of which at most four are normal, and cyclic incidence structures not meeting the budget of item 6; B1 when the maximal block containing the hub core has $p \in \{9, 10\}$ and every one of its vertices starts an induced P_7.

Appendix D. Other refuted routes

Here I_j is the number of *oriented* induced paths on j vertices (each path counted once in each direction). The path-count recurrence $I_{2k+1} \geq (\ell - k) I_{2k-1}$ holds for $k = 1$ in the form $I_3 \geq 2(\ell - 1)n$, with equality for cycles (for unoriented counts it fails already for C_5), but fails for $k = 2$: the Petersen graph with a handle and 29 leaves ($n = 42$) has $I_3 = 940$, $I_5 = 220$ and $nI_5 - (S - 2n)I_3 = -160$. For the random induced walk, the start vertex is chosen uniformly and each step chooses uniformly among the neighbours of the current end that keep the path induced, until there is none; the proposed inequality $\mathbb{E}\lceil |P|/2 \rceil \geq \ell$ is tight on the Petersen graph but fails on the Robertson graph, where $\mathbb{E}\lceil |P|/2 \rceil = 71/18 < 4 = \ell$. Single-ball and two-ball deletion lemmas fail on the Heawood graph and on B_5 (Example 11.2), respectively. For the terminal correction TD, singleton deletion and geodesic amplification provably cannot work: Petersen amalgams with a tail attain TD with equality at $q = 5$ with the tip the unique cheap non-cut vertex, and C_4-safe geodesic amalgamation can increase the rooted maximum.

Appendix E. Further partial results

The lemmas marked proved have their hypotheses written in full and their proofs written below. The counterexample is a single explicit graph. Every other note is quarantined: its enumeration, or a certificate it depends on, is not in this archive, so it is not a theorem and it is not on the claim ladder. No note in this appendix is unrestricted GEN_6, and none is REAL-133. Write $\varepsilon_6(G) = S(G) - 6|G|$. A nonempty set is 6-cheap when its deletion loss is at most six times its order. A quotient is *proper* when the output has fewer vertices than the input.

Lemma E.1 (ports; proved)**.** *Let G be triangle-free and strong C_4-free, and let $D \subseteq V(G)$ be a connected dominating set such that $d_G(x, y) = d_{G[D]}(x, y)$ for all $x, y \in D$. Write $H = V(G) \setminus D$.*

1. *Every vertex of H has exactly one neighbour in D. Call that neighbour its* port.
2. *If $x, y \in H$ are adjacent, with ports a and b, then $a \neq b$, $a \not\sim b$, and $d_{G[D]}(a, b) \leq 3$.*

Proof. Let $x \in H$. Since D dominates and $x \notin D$, the vertex x has a neighbour in D. If $a, b \in N(x) \cap D$ are distinct, then $d_G(a, b) \leq 2$, hence $d_{G[D]}(a, b) \leq 2$ by the distance hypothesis. Distance 1 gives the triangle axb. Distance 2 gives a 4-cycle through x and a length-2 path in $G[D]$. This proves (1).

For (2), let $x \sim y$ with ports a and b. If $a = b$, then axy is a triangle. If $a \sim b$, then $axyb$ is a 4-cycle. Thus $a \neq b$ and $a \not\sim b$. The path a, x, y, b has length 3, so $d_G(a, b) \leq 3$, and the distance hypothesis gives the same bound in $G[D]$. □

Lemma E.2 (cut size; proved)**.** *Under the hypotheses of Lemma E.1, the cut $E(D,H)$ has exactly $|H|$ edges, and*

$$\varepsilon_6(G) = 2e(G[D]) - 6|D| + 2e(G[H]) - 4|H|.$$

If in addition $e(G[H]) \le 2|H|$, then $\varepsilon_6(G) \le 2e(G[D]) - 6|D|$. The inequality $\varepsilon_6(G) \le -2|D|$ does not follow from this identity alone.

Proof. Each vertex of H has exactly one neighbour in D, so the cut has $|H|$ edges. Lemma 2.1(3) gives $S(G) = 2\big(e(G[D]) + e(G[H]) + |H|\big)$. Subtract $6(|D| + |H|)$. □

Quarantined lemma E.3 (outside degree; not independently checkable from this archive)**.** *Let G be triangle-free and strong C_4-free, and let D and H be as in Lemma E.1, so every vertex of H has a unique port in D. Let T be a tree and let c be a vertex of T of degree $d \in \{1,2,3,4\}$. Write $N_2(c)$ for the set of vertices at distance exactly 2 from c in T.*

The following numerical sentence was compared with a list of 1,208 *trees of order at most* 16*. That list is not in this paper. The rule that selected the trees, a bound of* 4 *on the degrees of certain terminal vertices together with a separation condition between branch vertices, is not written here. The sentence is: $|N_2(c)| \le 3,4,3,1$ according as $d = 1,2,3,4$.*

A second sentence, using the unique-port property, asserts that if $G[H]$ arises under that unwritten selection rule then $\Delta(G[H]) \le 4$ and $e(G[H]) - 2|H| \le 0$. Lemma E.2 would then give $\varepsilon_6(G) \le 2e(G[D]) - 6|D|$. The stronger inequality $\varepsilon_6(G) \le -2|D|$ is not a consequence of Lemma E.2. Neither sentence is a theorem of this paper. Both apply only when a set D as in Lemma E.1 is given. They do not assert that every triangle-free strong C_4-free graph of path order 11 *or* 12 *has a nonempty* 6*-cheap set.*

If D is a single vertex and $\delta(G) \ge 4$, then Lemma E.1 forces G to be a star, whose leaves have degree 1. Thus no one-vertex set D satisfies the hypotheses of Lemma E.1 in a graph of minimum degree at least 4. This does not say that every graph of minimum degree at least 4 has some other set to which the quarantined degree bound applies.

Lemma E.4 (averaging; proved)**.** *Let $k \ge 1$ and let $w_1, \dots, w_k$ be real numbers. Then $\max_i w_i \ge k^{-1}\sum_i w_i$. If $w_i = 2d_i - 2$ for an integer $d_i \ge 1$, then $w_i \ge 0$. If a nonempty finite set is partitioned into k colour classes and each class is given a real weight, some class of maximum weight is nonempty.*

Proof. The maximum is at least the average. If $d_i \ge 1$ then $2d_i - 2 \ge 0$. A class of maximum weight in a partition of a nonempty set may be chosen nonempty. □

Three further colouring bounds, with nonnegative weights of the form $2d-2$ on the colour classes of a port geometry, are verdict records only. The geometry, the number of colours, and the numerical parameters are not defined in this paper, so those bounds are not stated as lemmas and are not theorems. None of them asserts that an arbitrary member of the class $\mathcal{F}$ of Section 1 has a positive proper model. They are not applied to a one-vertex tree.

Lemma E.5 (counterexample; one quotient is not proper)**.** *Let G_{41} be the graph with graph6 code*
`hHDG?C@?GC_??@??_?G?H?????G??G??C??H'CPA?PCCOAICP?MAGaBPGAO?HGA?OC_GG_HG?oc@G?s_SA??I@?_?@OI??Q@?g?BCA_O?EgGGG??CCE?????@@__??OKCC??EOOOO?W`.

Then G_{41} has 41 *vertices and* 124 *edges, it is triangle-free, and every pair of vertices has at most one common neighbour, so G_{41} is strong C_4-free. Hence $S(G_{41}) = 248$ and $\varepsilon_6(G_{41}) = 2$.*

Let T be any one-vertex subgraph of G_{41}. Consider a quotient rule which, on an induced tree T, deletes a set determined by three nonnegative quantities a, b and σ attached to T, and which deletes nothing when $a = b = \sigma = 0$. For a one-vertex tree those three quantities are zero, so the numerical side condition $a + b + \sigma < \varepsilon_6(G_{41})$ holds, and the output of the rule is G_{41} itself. The output has the same order as the input, so it is not proper.

The lemma does not assert that G_{41} admits no positive proper model by some other rule. An appeal to an order-40 mass bound for that stronger sentence is not a proof in this paper, and that bound is not adopted. A repaired rule that assumes $|T| \geq 2$ is not stated here and is not a theorem.

Quarantined lemma E.6 (cubic bridge leaves; not independently checkable from this archive)**.** *Let G be a finite triangle-free strong C_4-free graph with $\varepsilon_6(G) > 0$ and $p(G) \leq 12$. Let b and m be nonnegative integers with $b \leq 3m$. Suppose some bridge of G (an edge whose deletion increases the number of components) has a degree-3 end, that every vertex incident with the bridges under consideration has degree at most 4, and that b is the number of leaves of those bridges.*

The recorded conclusion is that deleting a perfect matching of those leaves changes the surplus accounting by at most $4b/3$, hence by at most $4m$, and that the resulting connected graph is proper, triangle-free and strong C_4-free, and still has positive 6-surplus. The matching and the quotient are not constructed in this paper. The record is relative to a round R45 certificate, which is not a file in this repository, and to rounds R50 and R51. The conclusion is not a theorem.

A separate recorded cost inequality says that for every integer $K \geq 3$, if d runs over the degrees of the deleted leaves and D is a fixed parameter of the same accounting, then $\sum(2d - K) \leq (2D - K)\, 2b/3$. Here D is that parameter, not the set of Lemma E.1. The same record keeps, at an unspecified original scope, an implication from GEN_{K-1} and two further unnamed inputs to GEN_K. Those inputs are not defined in this paper. The implication is not GEN_K at the cap $p \leq 2K$ for any $K \geq 7$, and it is not unrestricted GEN_6.

Quarantined lemma E.7 (cactus sentences; not independently checkable from this archive)**.** *No class of cactus graphs is defined in this appendix. The following three sentences are verdict records, not theorems. Their constructions are not given here.*

(1) *If a cycle of length at least 9 is chosen in a cactus and v lies off that cycle, then v lies on an induced path of order 10 that meets the cycle, and a mass bound for the blocks around the cycle is recorded. The mass bound is not written in this paper.*
(2) *In two end-block configurations, either an induced path of order 13 appears, or a core which is a star of a stated numerical type satisfies a local inequality. The inequality and the type are not written here.*
(3) *Let Y be a deleted pair of vertices, and let $F(Y)$ be the number of edges in a minimum cover of a bipartite conflict graph on Y. The record asserts $F(Y) \leq 6$ and asserts that after a contraction which deletes two edges and one triangle the surplus is $\varepsilon_6 + 5 - F(Y)$. The conflict graph is not constructed here.*

These sentences do not cover every 2-connected graph of path order at most 12 that contains a triangle. They are not unrestricted GEN_6.

Quarantined lemma E.8 (radius source; not independently checkable from this archive)**.** *Let $r \geq 11$. This lemma does not assert $S(G) \leq (r-1)|G|$ for an arbitrary connected strong C_4-free graph of radius r and path order $2r - 1$. A verdict record describes a subclass in which a core C is chosen by a peeling rule and the complement $V(G) \setminus C$ is a cheap set, subject to a numerical budget. The peeling rule and the budget are not written in this paper, so the subclass is not a defined family of this article and the sentence is not a theorem. The same record says that a complementary class is still present. REAL-133 is open.*

Statement	What is established in this paper	Mark
Lemmas E.1, E.2	unique ports; cut identity	proved
Lemma E.4	the maximum is at least the average	proved
Lemma E.5	one one-vertex quotient equals G_{41} and is not proper	counterexample
Quarantined lemma E.3	a degree bound whose tree list is absent	not a theorem
Quarantined lemma E.6	a bridge-leaf quotient; round R45 is absent	not a theorem
Quarantined lemmas E.7, E.8	sentences whose constructions are not written here	not theorems

None of the quarantined lemmas is a hypothesis of a proved theorem. In particular they do not imply unrestricted GEN_6, and Quarantined lemma E.3 does not produce a 6-cheap set in every endpoint-bounded triangle-free graph of path order 11 or 12.

Appendix F. Index of proofs, certificates and referee reports

The certificate files that this repository actually contains are listed below. Each SHA-256 value is the hash of the JSON bytes. The sidecar in `certificates/` repeats that hash in `sha256sum` form. Section 16 gives the check commands.

File under `certificates/`	SHA-256
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`certificates/ROUND_BRANCH_R2_CERTI FICATE.json`	`e1db58a751f33e1fb40dab786609b46146969b6e b1395fab5ebc14c7f515e46c`
`certificates/ROUND_BRANCH_R3_CERTI FICATE.json`	`cef23e9c94e16d64c647b7babbb792e54a62b7ab 3d2490c2782eb5bf3b362f2b`
`certificates/ROUND_BRANCH_R4_CERTI FICATE.json`	`42c860ade94c3037ee9d448bd0945c932780738a bda0588cc578c514959b8443`
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`certificates/ROUND_BRANCH_R10_CERT IFICATE.json`	`3fc0bd9179f01d6537428a52ce01e72baf3d5ae1 e0720d456f98e22b72687706`
`certificates/ROUND_BRANCH_R11_CERT IFICATE.json`	`279ffa80b7dad0c81c675184001d3136bcca95fe d6a4c527ccdc44eea1917d7c`
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`certificates/ROUND_BRANCH_R13_CERT IFICATE.json`	`65be08d6da4472882eeb1116c68ae877745a35ff 94d03e817a5f106a230622fa`
`certificates/ROUND_BRANCH_R14_CERT IFICATE.json`	`bf9c54e049f0e994c1f7c74f697098ab5f8577e1 4ab46a46751d281a2a4655a8`
`certificates/ROUND_BRANCH_R15_CERT IFICATE.json`	`b19d57690283b2e5fa5b90dad78ec5d890b365f1 520e9348c805a771b127b293`

File under `certificates/`	SHA-256
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certificates/ROUND_BRANCH_R17_CERT IFICATE.json	4fac5fa2984cf9fb528de431ff05391d72757ad0 08c988fc2342f5d7a05871e5
certificates/ROUND_BRANCH_R18_CERT IFICATE.json	eb251a5db4654ed820939b137f47c59623c3f0e5 ed7421bff9c115e373e780e8
certificates/ROUND_BRANCH_R19_CERT IFICATE.json	c64b5ef4000d56e19e0a8560af7a6ab7e3d89fb2 6796b64c04b23870bd0d616c
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certificates/ROUND_BRANCH_R21_CERT IFICATE.json	e890a775b9565a6b86e44f83b2393b84b6a3f225 2573cdfbfabc602db12e9075
certificates/ROUND_BRANCH_R22_CERT IFICATE.json	fa45115f68645a0393efe9e23a6608a49c612d85 983bac3c86a2e1dddb292170
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certificates/ROUND_BRANCH_R30_CERT IFICATE.json	3f3b111a9bcb7a9b11831d55e6a0786a85ff6755 1e620375b6c82d7627239bbb
certificates/ROUND_BRANCH_R31_CERT IFICATE.json	fa0b2e093e8721345a0bbe452949f547405182a6 515837bbfebbd451c8708764
certificates/ROUND_BRANCH_R32_CERT IFICATE.json	46340d86df9ee4792c9f4c799ed8c0308387d3cd ca17c4ec926134188e2e8a21
certificates/ROUND_BRANCH_R33_CERT IFICATE.json	ddd07bbaefb662739825062c0a536752d6c7b199 21dfc872999ef626e930d529
certificates/ROUND_BRANCH_R34_CERT IFICATE.json	8d29038b7bb95d90c21d2bea517dcb28302d6024 2367becb8d589c254fa3aa4f
certificates/ROUND_BRANCH_R35_CERT IFICATE.json	132f7ecaba1663f20811bb8930e71f55ac481aa5 5003d638ca2ad3c0841eb0af
certificates/ROUND_BRANCH_R36_CERT IFICATE.json	410e10c00646bc3cba711a601b826b83a2078c4f ec0b474ab0340e2c91137b34
certificates/ROUND_BRANCH_R37_CERT IFICATE.json	27894d51a027f033ec005b94f6ff57f75c1a4bdb 866eb926af207324b6683758
certificates/ROUND_BRANCH_R38_CERT IFICATE.json	0f199e4ec5681d92467b04bd0747029235edba33 41ef7b3f3f13feeec2ee2bd5
certificates/ROUND_BRANCH_R39_CERT IFICATE.json	dc1224bcc8843e6fdbcf285e68d036e0945165ac b585b1c8f65c5bec5bb34477
certificates/ROUND_BRANCH_R40_CERT IFICATE.json	71819fcdd5681a6cb5413a6ff326d1d6d651ebd5 f2e316886d8f4abc0de9e4f7
certificates/ROUND_BRANCH_R46_CERT IFICATE.json	18acca0d37550ed985cb9b5f8b205c16a22bdbc7 88d2084a52e940aba9daf030

File under certificates/	SHA-256
certificates/ROUND_BRANCH_R47_CERT IFICATE.json	da0ef7adb88d62efd4cacac7281c80bfeead1c81 e810b102e42e07253176aae7
certificates/ROUND_BRANCH_R48_CERT IFICATE.json	1ebbc46c67c63a1ecc4fad07a95aa325cc51687c 077b75109839072ef7d8dd4e
certificates/ROUND_BRANCH_R49_CERT IFICATE.json	e95aabd77cdf22b8669cd27c8943e8e51ec93da5 740255c7f8c363f4af41c44b
certificates/ROUND_BRANCH_R50_CERT IFICATE.json	6a1c7aece7bc53906548cfb9f896b9ea32480f51 d1edb9de46acbe693824b111
certificates/ROUND_BRANCH_R51_CERT IFICATE.json	aebe0c00de6fd8ed0059280a2ac6d9c56e6c6435 5487912d0fde5a5be04999f7
certificates/ROUND_BRANCH_R52_CERT IFICATE.json	8383d17dac2d9f23995b066b4a9e82a6b977602e e9451bab90fd810161ee171a
certificates/ROUND_BRANCH_R53_CERT IFICATE.json	b90cc797e755cb8cf849937c20ad1576ea551a7b 0ec3b780676c3e84d81b6ecd

The lane paths in the next table, of the form data/rounds/, lanes/ and verdicts/, were the layout of a working tree that is not part of this repository. They are citations of where a written argument was produced. They are not files a reader can open here. A result whose only detailed proof lives on such a path is qualified in Section 16.

Paths below are written as in that working tree. For each lane, the written proof was report.md and, where present, scratch/REPORT.md. Referee lanes contain the referees' own code, which is not included here.

Result	Proof and certificates	Referee / audit
Lemmas 5.6–5.8, Thm. 5.1	§5; L/astra_r1/A; verify_round_branch_r1.py; data/lean/	L/astra_r2/audit; audits_r2_r6/astra_r2_audit.report.md
Thm. 6.1	L/astra_r2/gen4; referee_data/GEN4_star_certificate_p9.json; verify_round_branch_r2.py	audits_r2_r6/astra_r3_audit4.report.md
Lemma 7.1, Thm. 7.2	§7.1; data/check_lemmas.py	–
Thm. 7.5	L/astra_r3/rad5/scratch/RAD5_PROOF.md	audits_r2_r6/astra_r4_audit5.report.md
Thm. 7.6, ROOT(4)	L/astra_r3b/radK2/scratch/RAD6_ROOT_REPORT.md	audits_r2_r6/astra_r4_audit6.report.md
Thm. 7.7, ROOT(5)	L/astra_r4/rad7/scratch/RAD7_ROOT5_REPORT.md; referee_data/r7/ROOT5_EQUALITY_PROOF_radpush.md	audits_r2_r6/astra_r5_auditR7.report.md; V/astra_r7v/v_radpush
Thm. A.2	L/astra_r5/radr	audits_r2_r6/astra_r6_auditStruct.report.md
ROOT(6), sharp RAD(8)	L/astra_r7/root6, L/astra_r7/radpush; referee_data/r7/P6_EQUALITY_REPAIR.md	V/astra_r7v/v_root6, v_radpush
ROOT(7), sharp RAD(9)	L/astra_r8/root7 (certificates near_P7.json, near_P8.json, near_P9.json, prefix_*.json, *_exclusions*.json, . . .)	V/astra_r8v/v_root7
ROOT(8), sharp RAD(10), ROOT(9) partial	L/astra_r9/root8b (eight proof certificates far3_parentcap14.json, near_dK_a7*.json, $K = 0, \ldots, 6$)	V/astra_r9v/v_root8b

Result	Proof and certificates	Referee / audit
Thm. B.3, Lemma B.1	`L/astra_r8/root8` (§§2–5; `far4_patches.json`, `child_seeds8.json`)	`V/astra_r8v/v_root8`
Thm. 8.2 equality; component equivalence; conditional RAD	§8; `L/astra_r9/unify` §§1,3	`V/astra_r9v/v_unify`
Thm. 9.1, TD in $\mathcal{B}$	§9; `L/astra_r10/td`; `referee_data/r9/cactus_proof_counterexample.json`	`V/astra_r10v/v_td`
Thm. 9.5, clique-sums	`L/astra_r8/cells` (`bipartite_p11_certificate.json`, `bipartite_proof.py`); `L/astra_r9/cellsA2`	`V/astra_r8v/v_cells`, `V/astra_r9v/v_cellsA2`
App. C, items 2, 3	`L/astra_r4/cells`, `L/astra_r5/cellsre`, `L/astra_r7/cellsA1A2B1` (fixed-core certificates `A_*_stars.json`, `A_*_finish.json`, ...)	`audits_r2_r6/astra_r5_cellsre.report.md`; `V/astra_r7v/v_cellsA1A2B1`
App. C, items 4–7	`L/astra_r8/cells`, `L/astra_r8/cellsA1B1`, `L/astra_r9/cellsA2`	`V/astra_r8v/v_cells`, `v_cellsA1B1`; `V/astra_r9v/v_cellsA2`
App. C, items 8, 9	`L/astra_r9/cellsA1B1b`	`V/astra_r9v/v_cellsA1B1b`
App. C, item 10	`L/astra_r8/sel8`, `L/astra_r8/ep8`, `L/astra_r9/sel9` (endpoint certificates `critical*.json`, `star_K5.json`); `L/astra_r10/cb`	`V/astra_r8v/v_sel8`, `v_ep8`; `V/astra_r9v/v_sel9`; `V/astra_r10v/v_cb`
Ex. 11.1, 11.6	§11; `L/astra_r5/radr` §10; `verify_round_branch_r3.py`	`audits_r2_r6/astra_r6_auditStruct.report.md`
Ex. 11.2	`L/astra_r3/genK`	`verify_round_branch_r3.py`
Ex. 11.3, $\mathrm{EP}^+(6)$	`L/astra_r4/ep`, `L/astra_r5/ep8`, `L/astra_r7/ep78b`	`audits_r2_r6/astra_r6_auditStruct.report.md`; `V/astra_r7v/v_ep78b`
Ex. 11.4, 11.5, Conj. 13.3	`L/astra_r10/cb`; `referee_data/r10/petersen_cb.json`	`V/astra_r10v/v_cb`
App. D	`L/astra_r4/idea*`, `L/astra_r5/ideas`, `L/astra_r10/td`	`verify_round_branch_r5.py`; `V/astra_r10v/v_td`
Proved reductions, §1	written in the text	on the claim ladder; do not empty $\mathcal{R}_G$
Records (M), (S), (P)	`certificates/ROUND_BRANCH_R35_CERTIFICATE.json`–`R40`; search payloads absent	quarantined; not theorems
Lems. E.1, E.2, E.4, E.5	proved or counterexample in App. E	on the claim ladder, at the quantifiers written there
Quarantined lems. E.3–E.8	R51–R53 verdicts only; R45 absent; constructions not inlined as proofs	not on the claim ladder

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